



Evaluation-Driven RAG Framework for Grounded Research Paper Analysis

*MAJOR PROJECT REPORT submitted in partial fulfillment of the requirements
for the Award of the Degree of*

BACHELOR OF TECHNOLOGY

In

INFORMATION TECHNOLOGY

By

MANASA KOLUSU (228W1A1229)

SREEKAR MOVVA (228W1A1259)

Under the Guidance of

Dr.M.Suneetha, Ph.D.,

Professor and Head



**DEPARTMENT OF INFORMATION TECHNOLOGY
V R SIDDHARTHA ENGINEERING COLLEGE
(AUTONOMOUS - AFFILIATED TO JNTU-K, KAKINADA)**

Approved by AICTE & Accredited by NBA

KANURU, VIJAYAWADA-520007

ACADEMIC YEAR

(2025-26)

V.R.SIDDHARTHA ENGINEERING COLLEGE

(Affiliated to JNTUK: Kakinada, Approved by AICTE, Autonomous)
(An ISO certified and NBA accredited institution)
Kanuru, Vijayawada – 520007



CERTIFICATE

This is to certify that this project report titled “**Evaluation-Driven RAG Framework for Grounded Research Paper Analysis**” is a bonafide record of work done by **MANASA KOLUSU (228W1A1229), SREEKAR MOVVA (228W1A1259)** under my guidance and supervision is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology, **V.R. Siddhartha Engineering College** (Autonomous under JNTUK) during the year 2025-26.

(Dr.M.Suneetha)

Professor &Head

Dept. of Information Technology

(Dr. M. Suneetha)

Professor &Head

Dept. of Information Technology

EXTERNAL EXAMINER SIGNATURE

Date of examination:

DEPARTMENT OF INFORMATION TECHNOLOGY

V.R.SIDDHARTHA ENGINEERING COLLEGE

PROJECT SUMMARY

S.No	Item	Description
1	Project Title	Evaluation-Driven RAG Framework for Grounded Research Paper Analysis
2	Student Names & Numbers	Manasa Kolusu (228W1A1229) Sreekar Movva (228W1A1259)
3	Name of The Guide	Dr.M.Suneetha
4	Research Group	Retrieval-Augmented Generation and Artificial Intelligence
5	Application Area	Research Paper Analysis
6	Aim of the Project	The aim of this project is to develop an evaluation-driven retrieval-augmented generation (RAG) system that processes research papers and generates context-aware responses by integrating document retrieval with large language models.
7	Project Outcomes	1. Improved accuracy and relevance in document-based question answering. 2. Reduced hallucination through context-grounded response generation.

Student Signatures

- 1.
- 2.

Signature of the Guide

ACKNOWLEDGEMENT

First and foremost, I sincerely salute our esteemed institution **V.R SIDDHARTHA ENGINEERING COLLEGE** for giving me this opportunity for fulfilling my project. I am grateful to our Principal **Dr. A.V.RATNA PRASAD**, for his encouragement and support all through the way of my project.

On the submission of this Project report, I would like to extend my honour to **Dr. M.Suneetha**, Head of the Department, IT for her constant motivation and support during the course of my work.

I feel glad to express my deep sense of gratefulness to my project guide **Dr. M.Suneetha, Ph. D., Dean R&D, Professor & Head of department IT** for **his/her** guidance and assistance in completing this project successfully.

I would also like to convey my sincere indebtedness to all faculty members, including supporting staff of the Department, friends and family members who bestowed their great effort and guidance at appropriate times without which it would have been very difficult on my part to finish the project work.

TABLE OF CONTENTS

ACKNOWLEDGEMENT	iii
LIST OF FIGURES	vi
LIST OF TABLES	vii
LIST OF EQUATIONS	viii
ABSTRACT	i x
CHAPTER-1 Introduction	
1.1 Origion of the Problem	01
1.2 Basic definitions and Background	02
1.3 Problem Statement with Objectives and Outcomes	05
1.4 Societal Imapact Of Praposed Work	06
1.5 Applications	07
1.6 Summary	08
CHAPTER-2 Review of Literature	
2.1 Description of Existing Systems	10
2.2 Summary of Literature Study	14
2.3 Software Requirement Specification	17
2.4 Summary	19
CHAPTER-3 Proposed Methodology	
3.1 Design Methodology	20
3.2 Description of Algorithms	25
3.3 Description of dataset	31
3.4 System Architecture	34
3.5 Summary	37
CHAPTER-4 Results and Observations	
4.1 Description of Results	38
4.2 Model Performance Comparision	55

4.3 Observations from work	60
4.4 Summary	62
CHAPTER-5 Conclusion and Future work	
5.1 Conclusion	63
5.2 Future Work	64
References	x
Appendix	xii

LIST OF FIGURES

Figure	Page
Fig 3.1: Pipeline Diagram	21
Fig 3.2: System Architecture Diagram	36
Fig 4.1: LULC Classification for 1994	39
Fig 4.2: LULC Classification for 2004	40
Fig 4.3: LULC Classification for 2014	41
Fig 4.4: LULC Classification for 2024	42
Fig 4.5: Accuracy Comparison	43
Fig 4.6: Max Temperature Analysis	44
Fig 4.7: Min Temperature Analysis	45
Fig 4.8: Precipitation Analysis	46
Fig 4.9: Actual vs Reconstructed results	47
Fig 4.10: LSTM Loss Curve	48
Fig 4.11: BERTScore	49
Fig 4.12: ROUGE Analysis	50
Fig 4.13: Combined Metric Analysis	51
Fig 4.14: LLM-as-a-judge	52
Fig 4.15: Consistency Analysis	53
Fig 4.16: Landing Page	54
Fig 4.17: Chat Box	55
Fig A1: RAG Brain	xiv
Fig A2: LLM-as-a-judge Evaluation	xv
Fig A3: Section Splitting Logic	xvi
Fig A4.: Source Citation Logic	xvii

LIST OF TABLES

Table Name	Page
Table 2.1: Literature Review	15
Table 3.1: Random Forest	26
Table 3.2: SVM	27
Table 3.3: LSTM	28
Table 3.4: Cosine Similarity	29
Table 3.5: MiniLM-L6-v2	30
Table 3.6: Llama 3.3 70b	31
Table 4.1: Overall System Performance	57
Table 4.2: Lexical and Semantic evaluation	58
Table 4.3: LLM-as-a-judge	59

LIST OF EQUATIONS

Equation	Brief Description
(1)	Image Processing
(2)	Hyperplane
(3)	Optimization Function
(4)	Forget gate
(5)	Input gate
(6)	Cell gate
(7)	Cosine Similarity

ABSTRACT

.The growing adoption of large language models in academic research has pointed out major disadvantages of out-of-date knowledge and generated content, especially in the context of recent or field-specific information. In order to solve these issues, the proposed project outlines an evaluation-based retrieval-augmented generation model that would be applicable to the analysis of research papers. The system combines semantic retrieval and controlled response generation to make sure that responses are based on real document contents. The PDF format of research papers is treated in a pipeline, which consists of text extraction, preprocessing, semantic chunking, and embedding generation, and the vectors are stored in a vector database to enable them to be retrieved effectively by similarity. Upon a query, the cosine similarity is used to retrieve the relevant parts of the documents and put them into the prompt of the Llama 3.3 70B model, guaranteeing contextually aware and factually sound answers. The framework is assessed on both quantitative and qualitative metrics, such as BERTScore, ROUGE-L and LLM-based assessment of relevance, completeness and groundedness. Experiment findings show that there is an evident advantage over baseline model, semantic alignment increases to 0.851, ROUGE-L scores to 0.196 and groundedness scores to 4.48, with a significant decrease in hallucinated responses. The system also includes a counter-mechanism that helps to avoid the generation of responses in case of the lack of relevant context, increasing reliability. In summary, the suggested framework is a scalable and efficient solution to precise and context-sensitive analysis of research papers, bringing the outputs of language models into alignment with real-life academic needs.

Keywords: Retrieval-Augmented Generation, Large Language Models, Semantic Search, BERTScore, ROUGE-L, Grounded-AI.

Chapter 1

Introduction

The chapter presents the basic principles and motivation of the proposed retrieval-augmented generation framework of analyzing research papers. As digital academic content rapidly increases, there has been a growing desire to have systems capable of efficiently extract, comprehend and produce correct information in vast amounts of research documents. The use of traditional methods does not tend to cope with long and complex texts, which results in such problems as the lack of full comprehension and misleading results. Recent developments with large language models have enhanced natural language processing options, but the models remain limited to static knowledge and can generate hallucinated or non-verified information. In order to overcome such issues, the given system is proposed to combine semantic retrieval with the controlled text generation to make sure that the response is based on the real document material. This chapter gives a background of the issue at hand, explains the weakness of the current approaches and justifies the necessity of a more dependable and context-sensitive solution. It further explains the purpose of the intended work and its applicability in practices like aiding academic research and smart document analysis.

1.1 Origin of the Problem :

The growth of digital information in the academic and industrial spheres has caused an extremely high growth in the volume of research materials in electronic formats. Articles in scholarly journals, technical reports and conference papers are being produced and saved in large numbers and they are usually in the PDF file format. In as much as this expansion opens up extensive scope of knowledge, there are also issues of time-consumption in finding, processing and harvesting pertinent information. To extract the useful information, researchers and students will tend to spend a lot of time going through various documents manually, slowing down the research process and lowering the productivity in general. [6]

The conventional information retrieval systems mainly depend on the search method based on key word, which is not effective to tell the real meaning and intent of the user requests. Such systems often provide high volumes of vaguely-related or irrelevant documents, and users have to manually sieve and interpret the output. Additionally, these systems do not answer directly and instead refer to documents, thus rendering the system inefficient and time

consuming. The more intricate and interdisciplinary the nature of research topics is, the more clear the limitations of the key-word-based methods. [2], [3]

The current developments in large language models have enhanced the capability of machines to comprehend and produce text similar to that of human beings. These models, however, work with already trained knowledge and can lead to false or forged information, a phenomenon that is often known as hallucination [7]. This is a crucial limitation in academic contexts where accuracy and reliability are crucial. Furthermore, it is not easy to deal with long and complicated research paper because of the context constraints and the inability to be coherent. All these problems highlight the critical necessity of a system that integrates semantic retrieval and controlled generation in offering accurate, contextual, and reliable answers based on the real document content. [1], [4]

1.2 BASIC DEFINITIONS AND BACKGROUND

1.2.1 Retrieval-Augmented Generation (RAG):

The Retrieval-Augmented Generation is a hybrid model that integrates information retrieval with text generation in order to generate a correct and context-sensitive response. The system does not memorize an answer beforehand, but first of all, it retrieves the information that is relevant within a collection of documents and generates answers with the help of a language model, based on the information that has been retrieved. This will aid in mitigating hallucination and enhance factual accuracy. RAG has a specific purpose when applied in research activities, in which the answers are not produced based on assumptions but are rather grounded on real documents.

1.2.2 Large Language Models (LLM):

Large Language Models are deep learning models that learn on massive amounts of text to comprehend and produce human-like language. They employ transformer architecture and attention mechanisms in these models to learn the relationships between words and sentences. Although LLMs can produce meaningful and fluent answers, their nature is constrained by fixed training data and can give wrong or obsolete information. In this project, retrieval mechanisms are applied in combination with LLMs to make sure that the generated responses are contextually relevant and accurate.

1.2.3 Text Embeddings:

Text embeddings are numerical vector representations of text data, which contain semantic information. Every word is translated into a high dimensional vector, and the system is able to compare meanings rather than focusing on the exact words. As part of the proposed system, document fragments and user queries are transformed into embeddings with a pre-trained model. Such embeddings allow semantic search through detecting similarities of queries and stored document content, despite the use of different words.

1.2.4 Vector Databases:

A vector database is a special purpose storage system that can store high-dimensional vectors effectively. In contrast to conventional databases, which depend on the exact matching, the searches in a vector database are the similarity based searches that use embeddings. The document embeddings are made in a vector database in this project and relevant content can be quickly retrieved. When a query is processed, its embedding is compared with stored vectors to find the most relevant document segments.

1.2.5 Cosine Similarity:

Cosine similarity is a mathematical tool that is employed to identify a similarity between two vectors. It computes the cosine of the angle between two vectors in more than three dimensions. A high similarity corresponds to values near 1 and low similarity corresponds to values near 0. The proposed system has cosine similarity as a means to compare query embedding and document embedding to retrieve the required information accurately on the basis of semantic meaning.

1.2.6 Evaluation Metrics:

In order to evaluate the performance of the proposed system of retrieval-augmented generation, the below metrics of evaluation are applied:

BERTScore

BERTScore is a semantic similarity index between the generated answer and the reference answer. It compares meaning instead of word-for-word matching using contextual embeddings by transformer models. This enables the system to determine whether the result yielded by the system carries the same message as the original material. Higher BERTScore

means more significant alignment and meaning responses are apt to be found. This measure is especially applicable in the case of natural language generation.

ROUGE-L

ROUGE-L confirms that the text generated has lexical similarity to the reference text based on the length of the longest common sequence. It quantifies the extent to which the original text structure and words are retained in the output that is generated. The ROUGE-L score increases with the degree of overlap with the reference answer. This measure can be helpful in the context of comprehending the degree of adherence to the anticipated text pattern of the generated response.

Groundedness

Groundedness is the degree to which the response that is generated is justified by the content of the retrieved documents. It ensures that the data in the output is the one directly obtained by the source documents and was not created by the model. The high groundedness score means that the response is truthful and factual. This measure is crucial to minimizing hallucination and having credibility in research-based uses.

Relevance Score

Relevance score is the analysis based on the how suitable the generated response answers the query of the user. It is used to determine how closely the answer is to the question and also to eliminate irrelevant and unrelated information. The relevance score will be a higher indication that the system can retrieve and use relevant context. This measure will keep the response narrow and helpful to the user.

Completeness Score

Completeness score evaluates the completeness of the generated response against all the significant details of the query. It assesses whether the essential information has been given without omitting out important facts. The higher the score, the more detailed answer the response gives. This measure assists in making sure that the output is not merely accurate but informative enough as well.

1.3 PROBLEM STATEMENT WITH OBJECTIVES AND OUTCOMES

The increased rate of expansion of the digital research documents in most areas has posed a great challenge in terms of the efficient extraction of meaningful and relevant information. The length of academic papers, technical reports and scholarly articles are normally lengthy and have complex structure where manual analysis is time-consuming and inefficient. Conventional information retrieval systems are largely based on key word techniques, which can in most cases be unable to display the actual meaning of the queries made by the user. Consequentially, users have to go through several documents manually to discover useful information and it makes them less productive and more effortful. [6]

Over the past few years, big language models have shown significant performance in the ability to comprehend and create natural language. These models however are constrained by the reliance on the pre-trained knowledge and can produce inaccurate or unsupported information particularly when information in a particular domain or that published recently is involved. This phenomenon, normally called hallucination, can decrease the credibility of such systems in research-based practice. Also, there is the issue of dealing with long documents because of context length limitations and the inability to be coherent when dealing with big text inputs. [7]

The current strategies mostly involve retrieving information or text generation in isolation, without combining both processes to create a single system. The consequence of this partitioning is incomplete solutions that either access appropriate documents without eliciting direct responses or they provide responses that are not based on real data. As such, one requires a system that can integrate semantic retrieval and controlled text generation and offer correct, context, and reliable answers. The proposed system will resolve these issues through the creation of an integrated system that will analyze research documents effectively, extract the pertinent information, as stipulated by semantic similarity and produce responses based on the source material. [5]

1.3.1 Objectives:

- To create a system that combines retrieval and generation methods of processing research documents and giving the correct response.
- To minimize hallucination in generated outputs by basing responses on retrieved document content.

- To enhance the semantic interpretation of user queries on embedding-based similarity techniques.
- To make sure that created responses are topical, contextual, and correspond to original documents.
- To measure the performance of the system in terms of both quantitative and qualitative assessment.

1.3.2 Outcomes:

- A dynamic mechanism that can retrieve and produce correct answers of research documents.
- Less frequent hallucinated or unsubstantiated information in outputs generated.
- Better semantic relevance of queries made by users and the content of retrieved document.
- Dependable and context sensitive answers that can be used in academia and research.
- Enhanced overall effectiveness in analyzing and understanding research documents

1.4 SOCIETAL IMPACT OF PROPOSED WORK

1. The suggested system allows effectively managing and analyzing vast amounts of digital documents, which are rapidly growing in many sectors. It also enables users to get valuable information that would not need to look through long documents to get the relevant information and create context-driven responses. This greatly enhances access to knowledge, minimizes information overload and aids in accelerated learning and research. It comes in handy more specifically to students and researchers who need rapid and accurate grasp of academic contents [4].

2. The system improves decision-making in important areas like health, finance, education and governance. It also allows professionals to understand the intricate information fast and make decisions because they give the correct and trustworthy answers based on the factual contents of the documents. This lowers the possibility of human error, saves time and overall efficiency in areas where precision is very important. Precise data can be obtained quite rapidly when it becomes accessible, which can result in enhanced results in the process of information-driven processes [19].

3. Automation of document understanding minimizes reliance on manual processing which is time consuming and has the tendency of inconsistencies. The system reduces human input in

document searching and analyzing, enhancing efficiency and productivity. It is also scalable and can enable organizations to deal with big amounts of data without notable workforce and resources expansion. This renders it appropriate in present-day digital processes in which efficiency is a fundamental tendency [17].

4. The system encourages the application of believable and credible artificial intelligence through lessening hallucination in the produced answers. The system makes the information supplied reliable and verifiable since the outputs are based on the content of the documents that have been retrieved. This is more so in the academic and research settings where the accuracy of information is paramount. It helps to create trust in AI-based systems and stimulate their use in the sphere of knowledge-intensive applications [7].

5. The proposed approach enables digital transformation because it enables smart document processing frameworks, which can be tailored to different fields. It may be integrated into the existing systems to enhance them on information retrieval and analysis. It is this flexibility that allows organizations to rejuvenate their processes, and increase their efficiency without necessarily modifying their systems. The system will then aid in creating intelligent and automated solutions to knowledge management [15].

1.5 APPLICATIONS

Relevant Research and Academic Analysis:

The proposed system can be effectively used in the academic environment to assist the students, researchers, and scholars in analyzing vast volumes of research papers. It enables one to retrieve the corresponding information automatically and gives concise and context-sensitive responses such that, users can have a clear comprehension of complex academic information within a limited amount of time. This would go a long way in helping in literature review systems, knowledge discovery and efficient extraction of information. The system saves time on the multiple document analysis process, which increases productivity and enhances the overall research process.

Business Document Management:

The system can be employed in organizational contexts to handle and examine different kinds of documents, including reports, emails, contracts, internal records. It is useful in accessing important information and creating relevant summaries, which can allow employees and managers to make more decisions in a reduced time and with more quality information. This

will add efficiency to document organization, minimize manual work and optimize operations. The system also facilitates improved knowledge management in organizations through enabling quick access to information that is relevant.

Law and Finance Documentation:

The system is very practical in dealing with complicated legal and financial documents, which in most cases are lengthy and detailed. It helps to find valuable clauses, detect essential financial information, etc. and create summaries that point out essential information. This saves the work of reviewing the documents manually and enhances analysis accuracy. This can be useful to legal professionals and financial analysts in that they can perceive documents faster thus making their compliance and decision-making steps easier.

Healthcare Documentation:

The system can be applied in the medical field to prepare medical documents, clinical reports, and research papers. It assists the medical workers to find the necessary information in the shortest possible time which allows them to make more accurate diagnosis, plan the treatment and take care of the patient.

The system facilitates effective healthcare functions by minimizing the amount of time incurred in the process of going through huge amounts of documentation. It also establishes that the important information is not missed which is important in making medical decisions.

Knowledge-Based systems and Digital Assistants:

The suggested system can be incorporated to intelligent assistants and knowledge-based platforms with the purpose to give correct document-driven responses. It will supplement the capabilities of chatbots and artificial intelligence with the ability of the answers to be anchored in plausible sources of data. This would facilitate its implementation in learning systems, knowledge management systems and in customer support. Such type of integration improves user experience, because it provides precise and contextually-driven information in real-time.

1.6 SUMMARY

The present chapter gave the background and incentive behind the proposed retrieval-augmented generation framework to analyze research documents. It talked about the proliferation of digital documents and issues related to how to handle huge, complicated and multi-page research papers with a high level of efficiency. Time-consumption, Information overload, and extracting meaningful insights were found to be significant weaknesses of conventional document processing methods. The chapter has also discussed the currently available tools and their limitations especially when it comes to working with long documents, retaining a sense of contextual knowledge and the validity of the responses that are generated. The weaknesses of the standalone large language models such as the possibility of hallucination and dependence on static knowledge were also highlighted. To surmount these problems, the necessity of a comprehensive method that integrates semantic retrieval and driven text generation was determined.

CHAPTER – 2

REVIEW OF LITERATURE

The swift expansion of digital information over the recent years has resulted in the need to have intelligent systems that can process and interpret the vast amounts of unstructured data including PDF documents, research papers and reports. Conventional information retrieval techniques which rely on the matching of key words in information do not often provide any meaningful information about such data and this is particularly true in cases where one needs to accomplish or learn about the context of this information. As the Large Language Models (LLMs) and other deep learning approaches continue to evolve, the trend has shifted towards more intelligent and context-aware document analysis systems.

The chapter shows a systematic overview of the currently existing literature about document understanding, Retrieval- Augmented Generation (RAG) and hybrid forms of retrieval. It aims at examining various methodologies, models, and frameworks suggested by researchers to enhance the accuracy, efficiency and reliability of document-based question answering and summarization systems. The strengths and limitations of these approaches are also brought out in the chapter as a basis of discovering research gaps and justifying the necessity of the proposed system

2.1 DESCRIPTION OF EXISTING SYSTEM

Paper-1 Legal AI for All: Reducing Perplexity and Boosting Accuracy in Normative Texts With Fine-Tuned LLMs and RAG

Authors: PAULINA VIZCAÍNO, IVÁN GALO REYES-CHACÓN , PATRICIO SANTIAGO GARCÍA-MONTERO, MANUEL EUGENIO MOROCHO-CAYAMCELA

Year: 2025

Observations:

This paper focuses on improving the performance of Large Language Models in the legal domain by integrating fine-tuning techniques with Retrieval-Augmented Generation (RAG). The proposed approach enhances the ability of models to process complex normative texts by reducing perplexity and increasing response accuracy.

The system combines domain-specific fine-tuning with retrieval mechanisms to ensure that generated outputs are grounded in relevant legal documents. Experimental results show significant improvements in accuracy and reliability compared to baseline models [1].

Paper-2 Evaluation of Vector Database and LLM Models in Retrieval-Augmented Generation (RAG) Systems

Authors: Hilal Tekgoz, Halit Erdogan, Harun Uz, Hatice Altınok

Year: 2025

Observations:

This essay assesses the effectiveness of various vector databases and Large Language Models in the effectiveness of Retrieval-Augmented Generation systems. It evaluates different configurations to find out the effects of storage, indexing and retrieval mechanisms on the quality of response generated. The research notes that effective vector databases can greatly enhance speed and relevance of retrievals whereas the quality and coherence of responses depends on the selected LLM. The findings reveal that there are optimum combinations of model selection and vectors storage that could maximize the performance of the system though there are still challenges in balancing the latency, accuracy, and scalability [2].

Paper-3 QUIM-RAG: Advancing Retrieval-Augmented Generation With Inverted Question Matching for Enhanced QA Performance

Authors: BINITA SAHA , UTSHA SAHA, MUHAMMAD ZUBAIR MALIK

Year: 2025

Observations:

In this paper, presents QUIM-RAG a new method that improves Retrieval-Augmented Generation systems and applies inverted question matching methods. The technique enhances better comprehension of queries by restructuring user queries into several semantically equivalent ones that allow retrieval of information more effectively. This translates to better question-answering performance, especially in complicated query situations. Through experimental tests, it is seen that, QUIM-RAG is more accurate and relevant compared to traditional RAG models. Nonetheless, the method adds an extra processing cost since it involves several query transformations [3].

Paper-4 AI-driven Solution for Analyzing and Interpreting Data from PDFs Using RAG and Hybrid Retrieval Models

Authors: Dr. Granty Regina Elwin J, Rithik Raj K S, Raj Narayanan B, Moumitha R

Year: 2025

Observations:

This paper presents an AI-driven framework for extracting and analyzing information from PDF documents using Retrieval-Augmented Generation and hybrid retrieval techniques. The system combines semantic search with keyword-based methods to improve the accuracy of document understanding. It effectively handles unstructured data such as research papers and reports by segmenting documents and retrieving relevant content for response generation. The results indicate improved performance in information extraction and query answering. However, challenges remain in handling highly complex document layouts and ensuring consistent retrieval quality [4].

Paper-5 Dynamic Query Handling with RAG Fusion for PDF-Based Knowledge Retrieval Systems

Authors: Aakash Sonkar, Aayush Sahu, Sumeet Pratap Singh, Sachin Mishra, Komendra Sahu

Year: 2025

Observations:

This paper proposes a RAG Fusion-based approach for handling dynamic queries in PDF-based knowledge retrieval systems. The method enhances retrieval by combining results from multiple query variations, improving the diversity and relevance of retrieved information. This leads to more accurate and context-aware responses, especially for ambiguous or complex queries. Experimental results show improved retrieval performance and answer quality compared to standard RAG systems. However, the fusion process increases computational cost and may affect system efficiency [5].

Paper-6 The State of the Art in Conversational Document Analysis: A Review of RAG-Based Architectures

Authors: Hrishikesh Shiralaskar, Sachin Dagadkar, Ketan Wanjari, Sushil S. Chavhan, Tejas Gawande, Himanshu Chamatkar

Year: 2025

Observations:

This paper provides a comprehensive review of existing RAG-based architectures for conversational document analysis. It examines various system designs, retrieval techniques, and generation strategies used in modern document understanding systems. The study highlights key advancements such as hybrid retrieval, multi-document reasoning, and context-aware generation. It also identifies common challenges including hallucination, scalability, and evaluation inconsistencies. The paper serves as a foundational reference for understanding current trends and research gaps in RAG-based system [6].

Paper-7 Evaluating Hallucinations in Large Language Models (LLM): Metrics and Mitigation

Authors: Thrilok Kolla, Tina Babu, Rekha R Nair

Year: 2025

Observations:

This paper investigates the issue of hallucination in Large Language Models and proposes methods for evaluating and mitigating it. It introduces various evaluation metrics to measure factual correctness and reliability of generated responses. The study also explores mitigation techniques such as grounding responses in external knowledge sources and improving model alignment. Results show that incorporating retrieval mechanisms significantly reduces hallucination. However, completely eliminating hallucination remains a challenge, especially in complex and open-domain scenarios [7].

Paper-8 Dual-Source Retrieval for Educational Chatbots: A Performance Evaluation of Semantic Embedding Models

Authors: Samra Shaikh, Aparna Halbe

Year: 2025

Observations:

This paper presents a dual-source Retrieval-Augmented Generation system that combines structured PDF data and unstructured web content to improve chatbot performance in educational applications. It evaluates multiple embedding models based on both accuracy and efficiency using metrics such as BERTScore, ROUGE-L, and latency. The results show that high-performance models achieve better semantic accuracy, while lightweight models offer improved efficiency. The study highlights the importance of selecting appropriate embedding models based on application requirements and data characteristics, although it does not fully address hallucination and complex reasoning challenges [8].

2.2 SUMMARY OF LITERATURE STUDY

Literature review has underscored the recent development of Retrieval-Augmented Generation (RAG) systems and their increasing role as a system in document understanding and question-answering systems. Multiple studies focus on applying semantic retrieval methods to Large Language Models with the goal of enhancing the correctness, relevance, and confidence of the generated answers. Models like hybrid retrieval models, query expansion models and embedding based similarity search have demonstrated a great success in processing complex and unstructured documents like PDFs. [3][4][5]

Optimization of system components, such as vector databases, embedding models and query processing strategies are also the concern of many studies in an attempt to result in better performance and efficiency. Such techniques as multi-query retrieval, RAG fusion, and efficient embedding selection improve the performance of systems in terms of retrieving relevant information, particularly when complex or ambiguous queries are given. Also, domain-specific modifications, like legal document processing, show that fine-tuned models can achieve high accuracy when used in specialized tasks together with retrieval mechanisms. [1][2][8].

One such problem revealed in several studies is that hallucination in Large Language Models could occur, where the responses generated by it might not be consistently based on source information. Different measures of evaluation and mitigation techniques such as grounding, through retrieval and enhanced model alignment have been suggested to overcome this issue.

Though such techniques decrease hallucination, it is not easy to eradicate it completely, especially in open domain applications. [7]

Moreover, surveying research gives an overall view of the current RAG architectures, revealing progressive improvements in conversational document analysis and the most significant limitations: scalability, inefficient retrieval, and the absence of standard evaluation frameworks. Such lessons are used to comprehend the larger research context and new trends in document understanding systems. [6]

Comprehensively, the literature shows that despite the significant developments to date, there are still issues in providing reliability, scalability and efficient processing of long and complex documents. These loopholes underline the necessity of a highly developed and assessment-based RAG framework capable of offering precise, context-oriented, and credible answers that is the cornerstone of the proposed system. [1–8].

Table 2.1 Literature Review

Sno	Title and publication details	Algorithms or technologies used	Dataset	Purpose	Remarks
1	Year: 2025 Legal AI for All: Reducing Perplexity and Boosting Accuracy in Normative Texts With Fine-Tuned LLMs and RAG	Fine-tuned Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Domain-specific adaptation	Normative legal text datasets	To improve accuracy and reduce perplexity in legal document understanding	Enhances domain-specific performance but depends on high-quality labeled data
2	Year: 2025 Evaluation of Vector Database and LLM Models in Retrieval-Augmented Generation (RAG) Systems.	Vector Databases (Chroma, FAISS), Embedding models, Large Language Models	General question-answering datasets	To evaluate the impact of vector databases and LLM combinations on RAG performance	Shows trade-off between retrieval speed and response accuracy

3	Year: 2025 QuIM-RAG: Advancing Retrieval-Augmented Generation With Inverted Question Matching for Enhanced QA Performance	Retrieval-Augmented Generation (RAG), Inverted Question Matching, Query Reformulation	Question-answering benchmark datasets	To improve retrieval accuracy using query transformation techniques	Improves QA performance but increases computational overhead
4	Year: 2025 AI-driven Solution for Analyzing and Interpreting Data from PDFs Using RAG and Hybrid Retrieval Models	Retrieval-Augmented Generation (RAG), Hybrid Retrieval (semantic + keyword), PDF processing	Unstructured PDF documents	To extract and interpret information from PDF documents	Effective for unstructured data but limited in handling complex layouts
5	Year: 2025 Dynamic Query Handling with RAG Fusion for PDF-Based Knowledge Retrieval Systems	RAG Fusion, Multi-query retrieval, Query expansion	PDF-based knowledge datasets	To enhance retrieval using multiple query variations	Improves retrieval quality but increases processing cost
6	Year: 2025 The State of the Art in Conversational Document Analysis: A Review of RAG-Based Architectures	RAG architectures, Conversational AI, Hybrid retrieval techniques	Survey-based (multiple research datasets)	To review and analyze existing RAG-based document systems	Identifies research gaps and limitations in current systems
7	Year: 2025 Evaluating Hallucinations in Large Language Models (LLM): Metrics and Mitigation	Evaluation metrics, Grounding techniques, LLM alignment	Open-domain datasets	To evaluate and reduce hallucination in LLM outputs	Reduces hallucination but cannot eliminate it completely

8	Year: 2025 Dual-Source Retrieval for Educational Chatbots: A Performance Evaluation of Semantic Embedding Models	Retrieval-Augmented Generation (RAG), Embedding models, Dual-source retrieval	Structured PDF data and unstructured web data	To improve chatbot performance using multiple data sources	Balances accuracy and efficiency but lacks advanced reasoning capabilities
---	--	---	---	--	--

2.3 SOFTWARE REQUIREMENT SPECIFICATION

2.3.1 Purpose

This project aims to create an Evaluation-based Retrieval-Augmented Generation (RAG) system to smartly analyze research papers. The system is also designed to overcome the inadequacies of the traditional large language models such as outdated knowledge and hallucination due to response based on verified source documents. It is programmed to assist users in deriving quality, meaningful and contextual information in scholarly texts with a minimum hand-over.

The suggested system unites semantic search, vector databases, and transformer-based language models to allow efficient understanding of documents and answering questions. It also instills the machinery of evaluation such as the BERTScore, ROUGE-L and the LLM based evaluation in order to ensure that generated responses are accurate, in-depth and factual. The system offers an integrated pipeline, which integrates retrieval, generation, and evaluation to enhance reliability and trust in research-based uses.

2.3.2 Scope

This system has got a web based interface that enables the user to add research articles (in PDF format) and also query the system through queries in natural language. The backend reads the documents uploaded and extracts the text, breaks it down into semantically valuable units, generates embeddings and lets it be stored in a vector database, so that it could be retrieved easily.

When a user enters a query, the system finds the most relevant document fragments with the help of semantic similarity algorithms and feeds them into a big language model in order to produce the correct answers. Moreover, the system judges the resulting output by means of several measures to determine the presence of semantic accuracy, situational sensitivity and

factual basis.

The application offers features including document ingestion, semantic retrieval, generation of response using contexts, and performance evaluation. It can be applied in various fields such as academic research, technical documentation as well as knowledge management systems. The system is set to be scalable and efficient, which means that it will be able to deal with several documents and users and remain highly efficient and precise.

2.3.3 Intended Audience

- Students and researchers who require precise and fast facts in research papers.
- Specialists in technical and scholarly documents.
- Organizations that operate extensive knowledge stores.
- Population: Developers and data scientists working on AI-based document analysis systems.
- Students and teachers academic and understandings through AI-based tools.

2.3.4 Functional Requirements

- **FR1:** The system will enable the users to post research papers (in PDF format) to get them analyzed.
- **FR2:** The system will extract and pre-process text on uploaded documents including cleaning and breaking it down into smaller sections.
- **FR3:** The system will use document chunks, creating embeddings of vectors in a database of vectors to access these vectors easily.
- **FR4:** The system will access document parts that are relevant to queries by users through semantic search mechanisms.
- **FR5:** The system will be capable of producing answers based on a big language model based on the retrieved content.
- **FR6:** The system will test generated responses based on metrics of BERTScore, ROUGE-L and LLM-based evaluation.
- **FR7:** The system will show responses and the appropriate contextual information to bring about transparency.
- **FR8:** The system will be capable of managing a variety of queries and documents in one workflow.
- **FR9:** The system will not produce answers when there is inadequate relevant context.

2.3.5 Non-Functional Requirements

- **NFR1:** The system responding time will be optimized to give responses within a decent time (5-15 seconds depending on the complexity of query).
- **NFR2:** The system will be highly accurate and factual in the responses generated.
- **NFR3:** The system will be able to be scaled to support large document sets and multiple simultaneous users.
- **NFR4:** The system will be reliable and consistent to the various types of research documents.
- **NFR5:** The interface will be user-friendly, easy to navigate and intuitive.
- **NFR6:** The system will have data security and safeguard uploaded documents in processing.

2.4 SUMMARY

The chapter has provided an extensive overview of the literature on current studies in the field of document understanding, retrieval-augmented generation (RAG), and large language models. The different methods suggested by researchers were evaluated with a view of how contemporary systems process complex documents, enhance accuracy of retrieval and are able to produce meaningful responses.

Another finding highlighted by the review is that it is necessary to give more relevance to semantic search methodology and its integration with language modeling to enhance the performance of document based question answering systems in a broader sense.

There were many studies discussed in the chapter that conceived different aspects, including: the use of vectors databases, embedding models, multimodal document understanding and hybrid retrieval mechanisms. They carried out the finding that methods based on retrieval using generative models are much more relevant and accurate than the language models alone. They found that query comprehension and context retrieval were effective with techniques such as inverted question matching, dual-source retrieval and RAG fusion with enhanced performance in real life situation.

CHAPTER-3

PROPOSED METHODOLOGY

The chapter presents a proposed technique of developing an evaluation-based Retrieval-Augmented Generation (RAG) to examine research articles. The strategy is mindful about the cohesion of document processing, semantic retrieval, response generation and evaluation. The aim is to make sure that the system delivers the correct, contextualized and fact-based responses based on the relevant source documents.

The chapter describes the overall workflow of the system, which involves the ingestion of the documents, text, chunking, embedding generation, and the retrieval based on vectors. It also explains how a large language model makes use of the information collected to generate response and in addition assessment mechanisms to assess the quality of the response. This is a systematic method which assures greater reliability, less hallucination and enhanced performance when working in complex research documents.

3.1 DESIGN METHODOLOGY

This part introduces the architecture of the designed system and as such, the architecture is founded on an assessment-based Retrieval-Augmented Generation (RAG) framework of intelligent document comprehension and question answering. The system has been configured to effectively handle large volumes of research documents and produce precise and context-sensitive answers through combination of semantic retrieval and the use of large language models. In contrast to the conventional document processing systems based on the use of key-word match or isolated models, the suggested solution will make the generated outputs to be based on real document contents, which enhances the factual accuracy of the output and minimizes hallucination.

The methodology is a two-stage pipeline of Knowledge Base Creation, and Query-Based Answer Generation. The first stage involves converting raw documents into semantically meaningful vectors as a form of representation, which facilitates storage and retrieval of the data. The second phase involves processing user queries to find the proper information in the knowledge base and provide a correct response based on a large language model. The scalability, flexibility and performance of this design are enhanced by this and make the system functional and capable of handling multi-page documents.

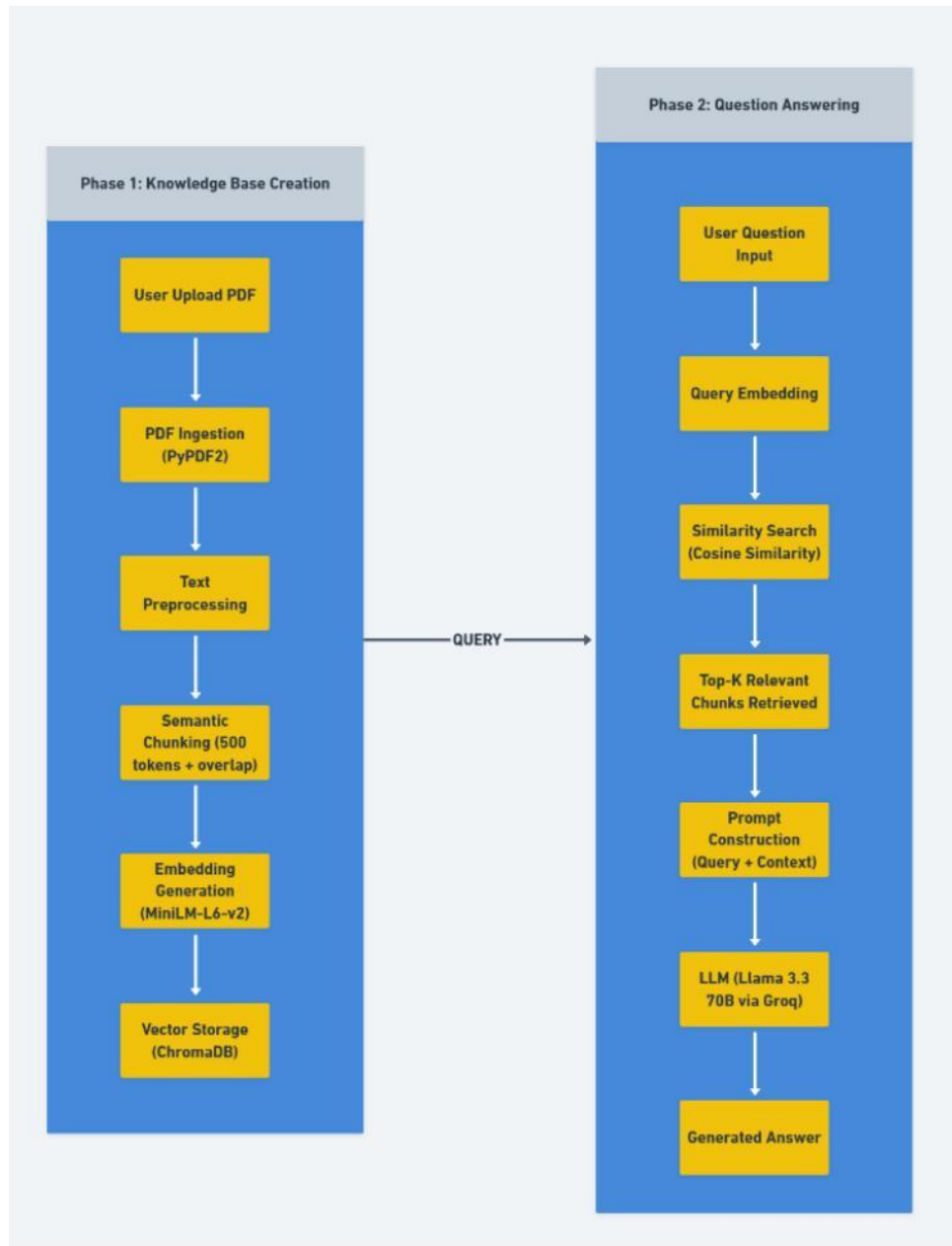


Fig 3.1 Pipeline Diagram

Phase 1: Creation of Knowledge base.

User Upload PDF:

The system interface as shown in Fig 3.1, allows the user to start the workflow by uploading a document in a PDF format. The documents can be research papers, reports, or technical documents that consist of both structured and unstructured content including headings, paragraphs, tables and figures. This is the initial step towards the creation of the knowledge base of the system. The system will be efficient in dealing multi-pages documents and help to

ensure that high performance does not decline by a significant margin when dealing with large files.

PDF Ingestion (PyPDF2):

After uploading the document, the system uses the PyPDF2 library to obtain the textual data of the PDF. This is done by parsing the document and transforming it into raw text format. As PDFs can be highly formatted documents, the objective of this step is to isolate readable and useful textual data and discard any non-essential formatting features. The ingestion module is very important in making sure that the extracted content is in an accurate form as the original document structure.

Text Preprocessing:

The data then undergoes a pre-processing step to improve the quality of data. This includes the removal of undesirable characters, special characters and inconsistency in format. The normalization tasks such as transforming the text to a consistent case and deleting of well meaning spaces can also be handled by the system. The step will also ensure the text has been cleaned, formatted and could be further processed thereby ensuring improved performance of the downstream tasks such as generating chunks and embedding.

Semantic Chunking (Overlapping 500 Tokens):

The processed text is cut into small semantic units of about 500 tokens in order to process long documents. A buffer of contextual continuity over the segments is ensured by maintaining an overlap between the recurring chunks. This is essential since, when text is split without overlapping, there can be loss of significant contextual relationships. Semantic chunking also makes sure that each segment maintains meaningful information and is still manages to make models.

Embedding Generation (MiniLM-L6-v2):

Every chunk is then converted to a vector representation with the MiniLM-L6-v2 embedding model. These embeddings represent the semantics of the text and not merely the lexical structure of the text. This enables the system to discern similarities among one or more parts of the text depending on sense as opposed to word-to-word comparisons. The lightweight but high-performance embedding models guarantee the balance between the performance and the computational efficiency.

Vector Storage (ChromaDB):

The results are the generated embeddings that are stored in a vector database in this case, ChromaDB. It is a database that serves as the knowledge base of the system, thus allowing high-dimensional vectors to be easily stored and retrieved. The use of vector storage enables the system to carry out quick similarity searches which is vital in query processing in real time. The system is scalable and can be used effectively because the database is capable of dealing with large amounts of data.

Phase 2: Question Answering

User Question Input:

The second phase as shown in Fig 3.1 Pipeline Diagram, is when the user communicates with the system, by making a query in a natural language. The system is configured to comprehend some complex queries which might need context and deduce information of several sections of the document. This process activates the process of responses and retrieval.

Query Embedding:

The query is then turned into a vector embedding by the exact same MiniLM-L6-v2 model that was used in the creation of the knowledge base phase. This guarantees consistency between query and document embeddings, which are then useful in making correct similarity comparisons. The system is able to understand user intent by formulating queries in a vector format.

Cosine Similarity (Similarity Search):

Similarity search in the system is done through cosine similarity to match query embedding with the document embeddings in the stored document vectors in the query database. This stage determines document fragments which are most closely related semantically to the query.

Top-K Relevant Chunks Retrieval:

The system retrieves the top-K most relevant document chunks basing on similarity scores. The most significant information needed to answer the query of the user is found in these chunks. The choice of multiple applicable chunks facilitates the capture of adequate context to the system; particularly to complex queries which involve several portions of a document.

Timely Construction (Query + Context):

The chunks retrieved are put together with the query of the user to make a structured prompt. This prompt provides contextual grounding to the language model which helps it generate accurate and relevant responses. Construction should be properly done in the right time, because it will directly affect the quality of the produced output.

LLM Processing (LLaMA 3.3 70b via Groq):

The generated prompt is fed through Groq to the LLaMa 3.3 70B model to generate a response. The mass language model works with the query and the retrieved context in order to produce a context sensitive and coherent answer. A high-capacity model is used that guarantees a high reasoning capacity and the quality of responses.

Generated Answer:

The user is given the final output of the model in the interface. Being founded on retrieved document content, the response is more trustworthy and factual than the purely generative ones. This is the final workflow in the system.

This ensures that the system will delivers clear, context-aware, and reliable responses to the user, therefore enhancing the overall usability and effectiveness of the application in real-world document analysis cases.

Overall Methodology Perspective

The methodology proposed according to Fig 3.1 unites process of document processing, semantic retrieval and language generation into one system. The system has higher modularity and scalability since it divides the workflow into two stages. Embeddings and the use of a vector database allow dealing with very large datasets, and a powerful language model is an element that ensures high quality results.

Moreover, similarity ranking techniques, and context-aware prompt construction are introduced into the system to increase the retrieval effectiveness and response accuracy. This method will greatly minimize hallucination and make the answers generated base on the real contents of the documents. Comprehensively, the design approach offers a powerful and extensible solution to intelligent document comprehension and question answering in a real-world.

In addition, the structure is designed to enable extensions in the future with better embedding models, better retrieval strategies and more efficient large language models as it evolves. This guarantees that since the concerns between data ingestion and the data in vectors are isolated, their storage, retrieval and response generation, the particular component can be optimised without necessarily affecting stability of the entire system. This enables maintainability and enables experimentation with other configurations such as hybrid retrieval techniques, reranking techniques as well as adaptive prompt engineering techniques. As a result, there is no prescribed implementation of the system but rather a naturalistic structure that can be adapted to different domains, document types and user requirements, which also underpins its irrelevance in large scale, real world intelligent information systems.

3.2 DESCRIPTION OF ALGORITHMS

1. Random Forest Algorithm

Random Forest is a machine learning algorithm, an ensemble algorithm applied to classification and regression problems. It operates by creating several decision trees in the training process and aggregating their results to enhance accuracy and minimize overfitting. The trees are constructed with random sub-sets of the data set and features, making the trees diverse. The make of the final prediction is through majority voting (to classify) or averaging (to regress). This is a method that enhances robustness and generalization performance than a single decision tree.

Random Forest is applied in LULC classification in the project as shown in Table 3.1 . The algorithm takes as input features of documents that are extracted and predicts the category by use of learned patterns. It assists in enhancing the accuracy of the classification because it minimizes the noise and complex distributions of data. It is ensemble in nature and can therefore be used in problems with structured datasets where different features are used to classify.

The high-dimensional feature spaces and the stability shown in random forest that work well under missing data or noisy data that random forest offers make it especially pertinent in the real world geospatial classification task. This helps to obtain more robust and stable LULC mapping outcomes in various temporal data sets.

Random Forest also has an in-built option of feature importance assessment, which enables determination of the most significant variables in the classification process. This is useful in determining which spectral bands or indices are most significant to LULC mapping. Therefore, it not only enhances model performance but also aids the enhanced interpretability of geospatial analysis findings.

Table 3.1 Random Forest

Algorithm:	Random Forest
Input:	Feature vectors extracted from document
Output:	Predicted document class
Method:	
1. Import dataset and preprocess features. 2. Select random subsets of data using bootstrapping. 3. Construct multiple decision trees using random feature selection. 4. Each tree independently predicts the output class. 5. Combine predictions using majority voting. 6. Final output = class with highest votes.	
$\hat{y} = o(1), o(2), \dots, o(n)$	

2. Support Vector machine (SVM).

Support Vector machine (SVM), an algorithm used in regression and classification, is a supervised learning algorithm. It works by finding an optimal hyperplane that separates data points of different classes with maximum margin. The algorithm concentrates on the support vectors, which are the most important data that are nearest to the decision boundary. Using SVM, it can work with both linear and non-linear data exposing the use of the linear, polynomial, and radial basis function (RBF) kernels. The SVM is very effective in working with high-dimensional data, and it is therefore appropriate to classify LULC with

multispectral satellite images. It has a high generalization ability despite the small training samples

The application of SVM in LULC Classification from Table 3.2, is to classify structured information and document

Table 3.2 SVM

Algorithm:	Support Vector Machine
Input:	Feature vectors
Output:	Class label
Method: 1. Map input data into high-dimensional feature space. 2. Identify optimal hyperplane separating classes. 3. Maximize margin between closest points (support vectors). 4. Apply kernel trick for non-linear classification. 5. Classify new input based on decision boundary. $() = + b \dots(2)$ $\min \frac{1}{2} ^2 \dots(3)$	

based features. It aids in determining definite classifications between classes and works excellently in high-dimensional spaces. The system can use the kernel functions to manage the intricate relationship among features maximizing classification performance.

3. Long Short-Term Memory (LSTM).

A Recurrent Neural Network (RNN) known as LSTM is meant to process sequence data and has the ability to capture temporal information. It involves memory cells and gating mechanisms (input gate, forget gate, output gate) to regulate the flow of information. This allows it to address the vanishing gradient problem of the traditional RNNs. Applications

LSTM finds extensive use in natural language processing applications like text prediction, sentiment analysis, and sequence modeling.

In Table 3.3, LSTM is applicable in analyzing text and processing sequential data. It aids in interpreting contextual relationships in text because it holds significant information across longer texts. This renders it useful in the context of language modeling and text classification where context is essential.

Table 3.3 LSTM

Algorithm:	Long Short-Term Memory (LSTM)
Input:	Sequential text data
Output:	Predicted sequence or classification
Method: 1. Input sequence is fed step-by-step into LSTM cells. 2. Apply forget gate to remove irrelevant information. $f_t = ([h_{t-1}, t] + b) \dots (4)$ 3. Apply input gate to store important information. $i_t = ([h_{t-1}, t] + b) \dots (5)$ 4. Update cell state. $c_t = c_{t-1} * f_t + i_t * \tilde{c}_t \dots (6)$ 5. Output gate produces final output. 6. Repeat for entire sequence.	

4. Cosine Similarity

Cosine Similarity is a mathematical value, which is applied to assess the similarity of two vectors with reference to the angle between two vectors. It has a wide usage in information retrieval and text analysis. Similarity is a value between -1 and 1 and 1 means the same

vectors, 0 means dissimilar. Cosine similarity is applied in the RAG Framework as you can see in Table 3.4 to compare the query embedding with document embeddings in the vector database. It assists in determining the most semantically legitimate document fragments that response to the user query as opposed to exact keyword.

Table 3.4 Cosine Similarity

Algorithm:	Cosine Similarity
Input:	Query vector, Document vectors
Output:	Similarity score
Method: 1. Convert query and documents into embeddings. 2. Compute dot product between vectors. 3. Calculate magnitude of vectors. 4. Apply cosine similarity formula. $similarity = \frac{v \cdot w}{ v \times w } \dots (7)$ 5. Rank documents based on similarity score.	

5. MiniLM (Embedding Model)

MiniLM is a model that is based on transformers and is utilized in generating semantic embeddings of text. It is able to capture contextual information and encode text to dense vectors. Such embeddings enable the efficient comparison of various text pieces in terms of similarity measures.

MiniLM-L6-v2 from Table 3.5 is employed to produce embeddings of document chunks and user queries in the project. This guarantees the presence of similarity search in that both are represented in the same vector space.

Table 3.5 MiniLM-L6-v2

Algorithm:	MiniLM-L6-v2
Input:	Text chunks / Query text
Output:	Dense vector embeddings
Method:	
1. Tokenize input text. 2. Pass tokens through transformer encoder layers. 3. Capture contextual relationships. 4. Generate fixed-length embeddings. 5. Store embeddings in vector database.	

6. LLaMA 3.3 70B (Large Language Model)

LLaMA 3.3 70B is a large-scale transformer-based language model, meant to be trained on advanced natural language generation and understanding. It applies self-attention to process the input text and respond to it in a context-sensitive manner. The model can work on large datasets and can perform complex reasoning.

The LLaMA 3.3 70B is employed in the last phase of the pipeline like seen in Table 3.6 in the project to produce answers depending on the user query and the context of the documents retrieved. It works on the constructed prompt (query + retrieved chunks) and generates a coherent and meaningful answer. This will make the output accurate and contextual.

The model also utilizes its scale pre-trained knowledge and contextual understanding patterns to produce fluent, logically structured and contextually consistent responses based on the clear retrieved information. Such effectiveness of recall context and generative ability offers a major enhancement in reliability and relevance of the end product.

Table 3.6 Llama 3.3 70b

Algorithm:	LLaMA 3.3 70B
Input:	Prompt (Query + Retrieved Context)
Output:	Generated answer
Method:	
1. Construct prompt using query and retrieved chunks. 2. Tokenize input prompt. 3. Pass tokens through transformer layers. 4. Apply self-attention mechanism. 5. Generate output sequence token-by-token. 6. Return final answer.	

3.3 DESCRIPTION OF DATASET

Dataset Source:

The data in this project is based on research orientated documents in publicly available academic and technical repositories including IEEE Xplore, arXiv, Springer and other open access digital libraries. The documents are mostly in PDF format and contain research papers, technical reports and scholarly articles in the field of computer science and artificial intelligence. The dataset is specifically selected to represent the complexity of real-world documents whereby each file can consist of several parts including abstract, introduction, methodology, results and references. Such a variety of document structure is a guarantee that the system will be tested in real-life conditions that require the ability not only to understand text but also to interpret the context. Also, the data includes files that have different writing styles, formatting patterns and domain-specific vocabulary, further complicating the processing and rendering it as an appropriate choice to test the strength of Retrieval-Augmented Generation systems.

Nature of Data:

The data is semi-structured as it is made of multi-page documents with structured and unstructured information. Both documents have a combination of plain text, headings, subheadings, tables, figures, captions and references, and the system will need to work with various types of content at the same time. This dataset has a number of challenges that need close preprocessing unlike simple text datasets like different font styles, different layouts, and inbuilt visual elements. The large-context dependencies required by the presence of long paragraphs and complicated sentence structure also require effective management. Such variation makes sure that the system will be able to generalize to various types of documents and will not be restricted to a particular format or a field. Subsequently, the dataset is instrumental in confirming that the system is able to execute the semantic understanding, retrieval and response generation processes successfully.

Data Content and Data format:

All the documents within the dataset exist in Portable Document format (PDF) which is popular in sharing of academic and professional contents. Text extraction libraries like Py PDF2 are used to extract the content of the PDF files and turn them into machine-readable text. To extract the information during the extraction process, the system would capture the textual information but leave out the non-textual features like images and sophisticated graphical features. The noise in the extracted content is usually in the form of unwarranted line breaks, special characters and format differences, which are processed by preprocessing. Cleaned text is the resulting input to the input steps of further processing. This step of PDF into structured text is a crucial step, as it will directly influence the quality of embeddings and retrieval in future steps of the system.

Preprocessing and Cleaning:

The data is first subjected to a whole preprocessing pipeline that aims at improving the quality and consistency of the data before it is used in the embedding and retrieval. Preprocessing phase involves normalization of the text, deletion of special characters, deletion of unnecessary whitespace and correction of encoding errors. Also, all the irrelevant information like page numbers, header and footers are filtered out so that only valuable information survives.

The use of tokenization to divide the text into manageable units is what helps to efficiently process using transformer-based models. The preprocessing step makes sure that the information that is fed into the system is clean, structured and semantically meaningful which enhances the overall performance of the retrieval and generation components.

Chunking Strategy:

The dataset is segmented into smaller parts with the help of a semantic chunking approach to deal with long documents effectively. Both documents are divided into tokens of about 500 tokens separated by a constant margin between adjacent portions to maintain the continuity of contexts. This redundancy allows critical information across the boundary of chunks to be preserved in segmentation. Not only does chunking make computations more efficient but also makes retrieval more accurate as the system is able to concentrate on smaller contextually relevant parts of the document. The long-document processing is especially crucial when it comes to the strategy since it allows the system to control memory in a way that semantic coherence is preserved.

Embedding Representation:

Every text chunk is then processed with the MiniLM-L6-v2 text embedding model to convert it into an entire text in a dense form. Such embeddings learn the semantic meaning of the text and enable the system to manipulate the similarity-based retrieval in an efficient way. The resulting vectors are indexed into a vector database like ChromaDB that can be indexed and retrieved at a high level. Since it represents text in a vector format, the system is able to compare the user queries to the stored document chunks based on mathematical similarity measures, and thus accurately and context-sensitive retrieved information can be obtained. This embedding-based method is at the heart of the Retrieval-Augmented Generation pipeline.

Scale and Size of data:

There are numerous research papers in the dataset with several pages of content, so the number of text chunks after the preprocessing and segmentation was large. The system will manage this increasing amount of data effectively through scalable storage and retrieval systems.

Relatability to the Project:

The selected data is highly relevant to the project aims, as it is highly similar to the real life scenario with complex and lengthy documents. It also enables the system to demonstrate that it can handle the long-context understanding, recall and generation of correct answers. The dataset supports in the evaluation of various system components, including preprocessing, embedding generation, retrieval accuracy, and response quality. Additionally, it offers a realistic testing conditions which may be used to determine the effectiveness of proposed RAG-based framework in minimizing the extent of hallucinations and maximizing factual accuracy. This renders the dataset a part of the overall verification of the system in terms of its performance and reliability.

On the whole, the data set employed in this project offers an excellent and realistic baseline on the assessment of the efficacy of the suggested Retrieval-Augmented Generation framework. It is diverse in terms of structure, content and formatting; this guarantees that the system is subjected to conditions that are close to real life cases. The preprocessing, semantic chunking and embedding representation allow the large scale and complex documents to be handled efficiently. This not only confirms the soundness of the retrieval and generation pipeline but also shows that the system can generalize to multiple types of documents and domains of types. Consequently, the dataset is vital in determining the reliability, scalability, and the useful applicability of the proposed methodology in intelligent document understanding and query answering problems.

3.4 SYSTEM ARCHITECTURE

The proposed model system architecture as shown in Fig 3.2 , is a modular and scalable Retrieval-Augmented Generation (RAG) framework that can combine document processing, semantic retrieval, and response generation into a single pipeline. The architecture starts with the user interfacing it via a frontend interface where he or she can upload documents and make queries in his or her own language. The frontend interacts with the backend API that serves as the central controller of the system. The backend process is tasked with the processing of requests, co-ordination of the various modules, and control of data flow through the pipeline. Based on the architecture, there are two major pipelines including the document ingestion and query processing pipeline whose culminating point is the vegetable database so that efficient matching can be made. This isolation would provide the ability to

have document processing and query handling work independently but remain synchronized by sharing storage.

Document ingestion pipeline as shown in Fig 3.2, converts raw PDF documents to structured and meaning representation that can be easily searched semantically. When a user uploads a document, it is initially run through a PDF pipeline with text being extracted in parsing. The result of the extraction is then taken through a chunking module where the document is broken down into smaller parts according to limits on tokens as context is preserved by way of overlapping.

This is then fed to an embedding model, namely MiniLM, which transforms each text fragment into a high-dimensional piece of data as a high-dimensional vector representing semantic meaning. The resulting embeddings are indexed into a vector database like ChromaDB that allows the storage and retrieval of the embeddings. This pipeline will make sure that big and complicated documents are converted to searchable entities so that the system can deal with long-contextual information.

In line with this, a query processing pipeline is invoked in a case where a user places a query via the frontend interface. The query is then transformed to a low-level representation as a vector with the same embedding model to ensure consistency between query and document representations. The embedding is then applied to conduct a similarity search in the vector database wherein the system recognizes the most relevant bits of the document due to their cosine similarity. The retriever module is important in the process of choosing the top-K relevant chunks that optimally respond to the query made by the user. This retrieval process will guarantee that it only transfers the most contextually holistic information to be subject to response generation, thus, minimizing noise and maximizing quality of answer. Embedding-based retrieval enables the system to go beyond the use of key words and to conduct profound semantic retrieval.

After retrieval of the corresponding chunks, they are sent to the prompt builder module, which builds a structured input, an amalgamation of the user query and the retrieved contextual information. This prompt can then be run on a large language model LLaMA 3.3 70B invoked in Groq which generates a coherent and context-sensitive response. A powerful transformer based model allows the system to generate not only accurate but also contextually based natural language answers that are based on the documents retrieved. The response generated is then relayed back to the frontend where it is presented to the user in a readable format. The architecture shown in Fig 3.2 in general focuses on performance,

scalability, and precision by integrating the use of vectors in retrieval along with sophisticated language models.

The modular design enables every part to be easily enhanced or replaced without compromising the functionality of the entire system. The retrieval is also quick with large datasets due to the use of vector databases in addition to the incorporation of retrieval and generation improving reliability of responses.

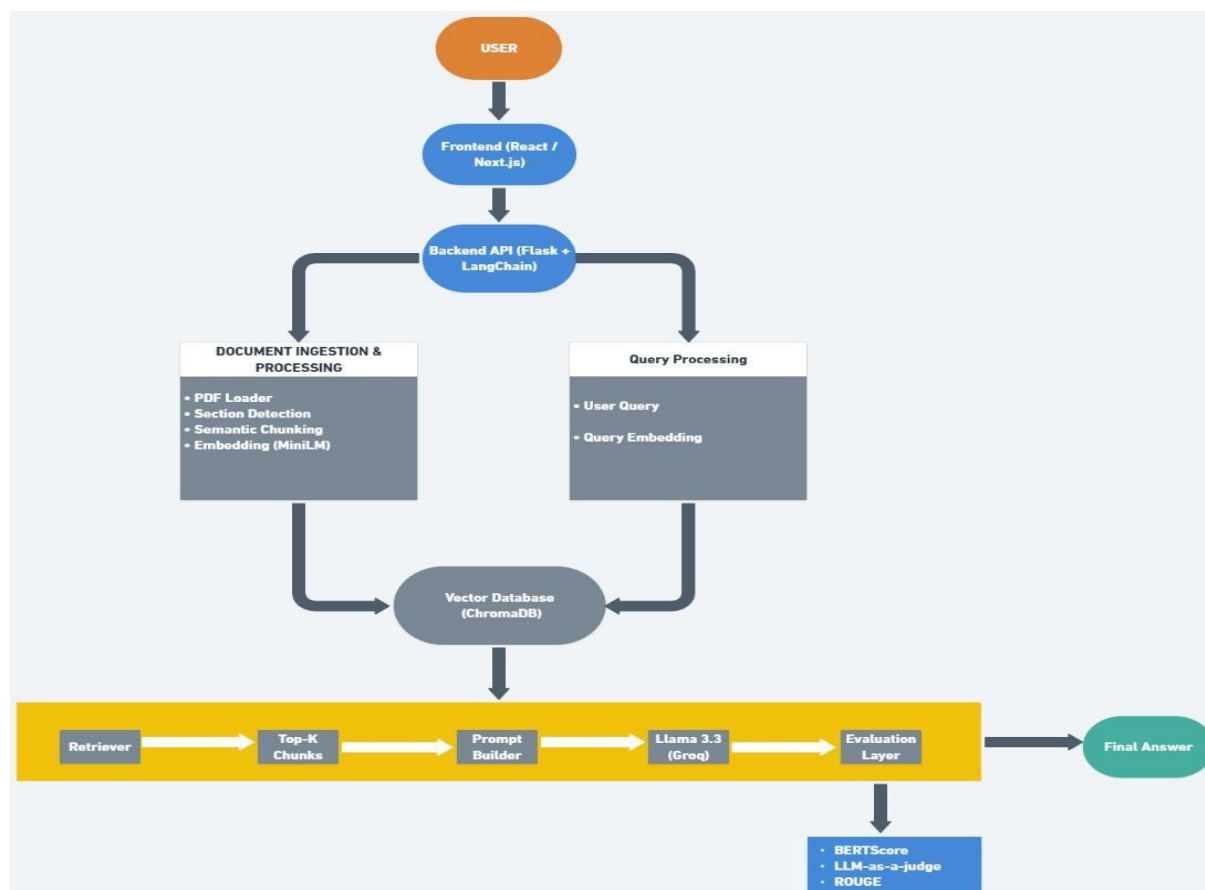


Fig 3.2 System Architecture Diagram

3.5 SUMMARY

This chapter presents the methodology of developing an evaluation-based Retrieval-Augmented Generation (RAG) system to understand the documents and generate the right answers. The approach incorporates key stages of document ingestion, preprocessing, semantic chunking, embedding generation, vector storage, retrieval and response generation utilizing large language models. Unlike the traditional systems, which are based on the application of key words, the proposed architecture involves the application of semantic retrievals and transformer based architecture to produce context based and fact based results. The system is effective in handling and handling bulk and complex documents transforming them into systematized representations of vectors that might be utilized to get access to information rapidly and precisely.

The methodology can be divided into two steps: creation of knowledge base and query-based response generation. Document processing is done to extract, clean, chunk and embed text which is stored in a vector database to be easily retrieved. The user queries are also incorporated and compared against each other on the basis of Cosine similarity and other techniques like the Maximum Marginal Relevance to maximize relevance and diversity. The system is built-in with several algorithms, such as Random Forest, SVM, LSTM, and LLaMA 3.3 70B, which are used in various tasks, such as classification, sequence modeling, and response generation. Overall, the modular architecture will ensure scalability and flexibility and increase accuracy consequently ensuring that the system is useful in dealing with complex and real-life documents with minimal hallucinations.

CHAPTER – 4

RESULTS AND OBSERVATIONS

In this chapter, the author provides an in-depth discussion of the findings of the realization of developed systems and estimates their performance regarding predetermined metrics. The research comprises the findings of LULC Classification and Climate Analysis, which are closely connected and are concerned with the same problem areas, which makes it possible to compare the two methods with each other. These two studies are assessed in order to comprehend how effective they are, what are their strengths, and what are their limitations as concerns the way they handle their particular tasks. A comparative study is conducted on Land use and Land cover Analysis and Climate Analysis to identify the aspects of improvement in terms of performance, accuracy, and efficiency, which gives an insight on the impact of various methodologies on the overall result.

Besides the these studies, the findings of the RAG Framework are also provided and discussed separately since it concerns another area of problems and is more developed and relies on the concept of Retrieval-Augmented Generation (RAG). The big project aims to comprehend the documents and answer the questions based on semantic retrieval and large language models and measure the performance of the big project with the help of semantic accuracy, consistency, redundancy, and information score. This project focuses on contextual knowledge and creation of meaningful responses to large and complex documents compared to these studies. This chapter gives in-depth observation of every system, identifying major findings, and performance gains, and practical implications of the suggested approach.

4.1 DESCRIPTION OF RESULTS

4.1.1 LULC Classification Results.

(a) LULC Classification Map -1994.

The 1994 Land Use and Land Cover (LULC) map of classification is the initial environmental status of the study area, which is used as a reference to study the long-term land changes. Preprocessed Landsat images were used to determine the classification as spectral signatures of various land cover types were identified and employed to train supervised machine learning models.

This map is essential because it will form the baseline of the first state that will be held against which the changes will be measured over time, and it will be very easy to comprehend the time-changes in land usage.

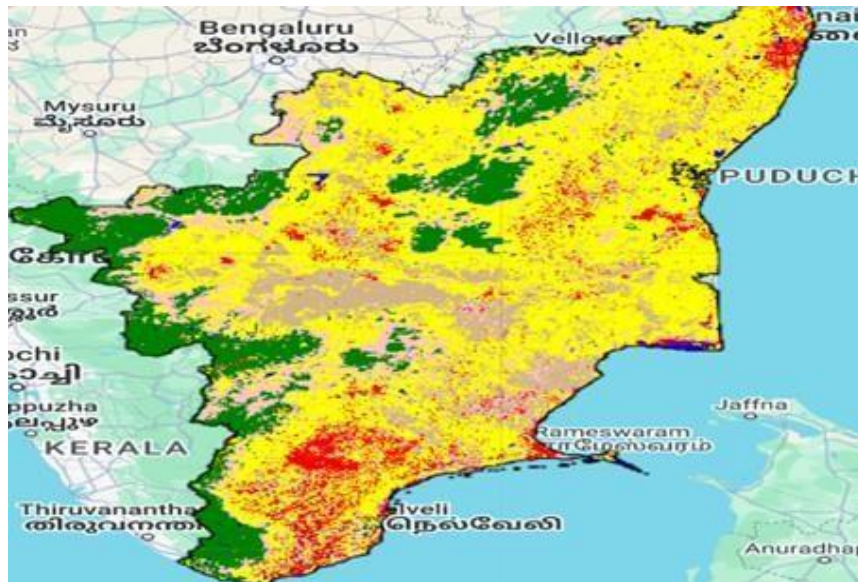


Fig 4.1 LULC Classification for 1994

The classification result of 1994 as shown in Fig 4.1, clearly indicates that agriculture was the major land use as it was complemented by the favorable environmental factors and less industrialization. The forest cover looks thick and is widely distributed which contributes to the ecological balance and biodiversity. Water bodies are also well recognized and this implies that there were stable hydrological conditions at this stage. The low concentration of constructed spaces indicates that urbanization was not so much an influence on the distribution of land. This minimally invasive map underscores a fairly sustainable land-use pattern, in which natural resources are conserved and human development is restrained. The sharpness of the separation of the classes in this map also speaks about the fact that this classification model works quite well with less complex and fragmented land patterns.

(b) LULC Classification Map -2004.

The 2004 LULC classification map is the mid-level stage of the land change, as it reflects the change that took place within 10 years period. This is the start of the era with the early signs of urban growth and the alteration of land use due to population growth and economic development. The classification process is the same as the prior stage making it comparable across time. Nevertheless, mixed land-use patterns do enhance spectral complexity of the data,

resulting in a more difficult classification process. This map is essential in determining the transitional changes and knowledge of how gradual changes in land use start to alter the entire landscape.

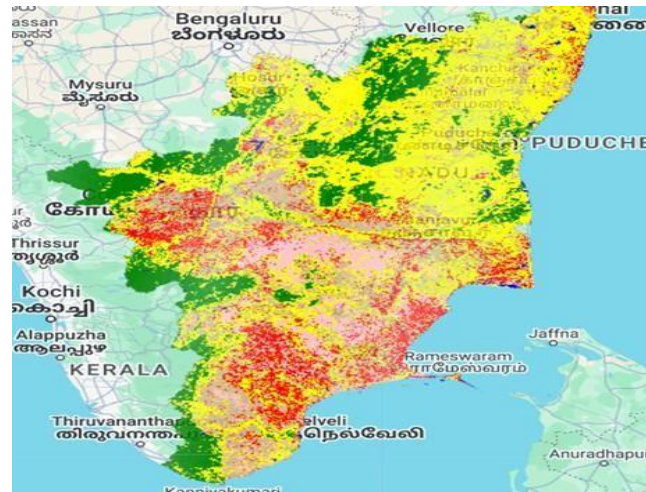


Fig 4.2 LULC Classification for 2004

The findings of the last classification in 2004 from Fig 4.2, show that the built-up areas are increasing significantly, especially in the vicinity of cities and rural developments. Farmlands start to become fragmented, parts of them turn into residential and commercial areas. In some regions, forest cover is marginally diminishing, which is an indicator of deforestation and conversion of land in their early developmental stages. Water bodies are very constant but can also have slight variations through seasonal and climatic conditions. The scenery looks more diverse in comparison with 1994 as it represents the onset of the human involvement in the land distribution. This phase shows the shift in the environment with mainly natural features to a more anthropocentric one.

(c) LULC Classification Map -2014.

The 2014 LULC classification map is a more developed phase of land transformations, with the impact of urbanization and environmental modifications being more evident. Development and expansion of human activities are very fast by this time leading to great impact on land-use patterns. The process of classification will have to contend with a larger spectral overlap caused by mixed land-use types and, as a result, the process of classification will also become more complicated. This map offers very useful information on the way the landscape is transformed during the process of longer-term urban development and pressure on the environment.

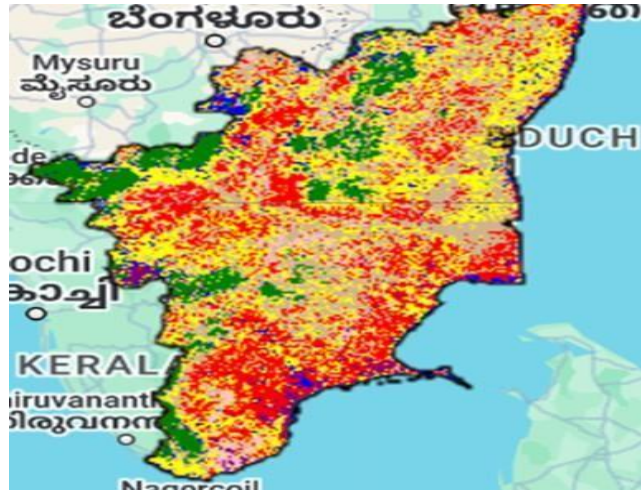


Fig 4.3 LULC Classification for 2014

The results of the classification in 2014 from Fig 4.3, show that the built-up areas have increased significantly, which means that urbanization and development of infrastructure are accelerating. There is still a reduction in agricultural land and the big part is transformed into non agricultural land. Deforestation and land encroachment are environmental challenges that can be observed in forest areas, as these areas are being noticeably degraded. The growth of barren or fallow land can be a sign of low agricultural productivity or desertion of land. The general impression of the landscape is, it looks much changed than it used to be in previous years as human activities have been the major determinant of land allocation. The fact that this stage shows a significant cumulative effect of development in 20 years.

(d) LULC Classification Map - 2024.

The 2024 LULC classification map (Fig 4.4) is the latest state of land use and will be used to give an overall picture of the long-term trends of changes observed in the past thirty years. The stage is the sum of the consequences of urbanization, industrialization, and environmental pollution. The classification model is used when using data that is very complex in nature in that the land-use categories are highly mixed and fragmented. The map will be the last step in the temporal analysis and will enable a full comparison of the years compared to the past and long-term patterns may be identified.

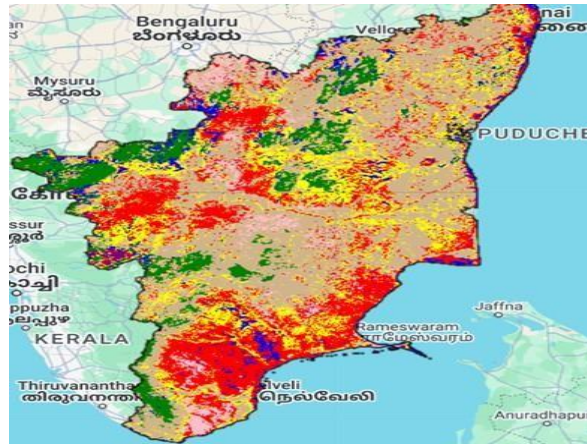


Fig 4.4 LULC Classification for 2024

The analysis of results in 2024 from the Fig 4.4 shows an obvious prevalence of developed areas, which proves the rapid growth of urbanization and the greater impact of humans on land use. The agricultural land area has gone down drastically, and this indicates the change in the land-use in the place of traditional land-use practices towards urban development. The forest cover has also decreased alarming ecological sustainability and biodiversity loss. The growth of arid and unproductive land points to environmental pressure and potential of soil quality deterioration. Climate variability as well as human intervention may also lead to the presence of changes in the water bodies. Generally, the map represents a massively changed landscape, in which natural environments are further giving way to urban and industrial complexes.

(e) The classification accuracy of SVM and random forest will be compared

Alongside spatial classification maps, overall accuracy and kappa coefficient are used as accuracy measures of machine learning algorithms. The analysis of Support Vector Machine (SVM) and Random Forest (RF) will give us information on how effective these models are in processing the data that is presented by remote sensing. This discussion is vital in identifying which model will be the most appropriate to employ in large-scale land classification.

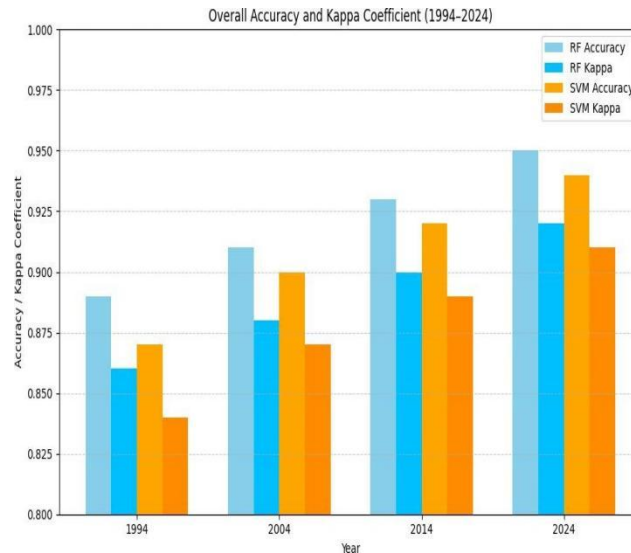


Fig 4.5 Accuracy Comparison

These findings in Fig 4.5, indicate that Random Forest is always more accurate and reliable at classification than SVM. Random Forest as an ensemble can minimize overfitting and optimally deal with complex data distributions because of its ensemble nature. SVM, though strong in some cases has been found to be weak in case of overlapping classes and heterogeneous land patterns. The better precision of the Random Forest shows that this algorithm can better be generalized to various land covers. The results confirm the application of ensemble learning methods to geospatial analysis and the significance of choosing suitable models to obtain a precise classification.

Also, the level of consistency in performances through the various land cover classes emphasizes the strength of Random Forest in managing real-world variabilities and complex environment. Its capability to preserve great accuracy despite noisy and overlapping data are especially useful when conducting large-scale geospatial analysis. This consistency guarantees that uniform classification outcomes are achieved when using various time databases, which enhances its capability in long-term LULC surveillance and decision-making.

4.1.2 Results of the Climate Analysis.

(a) Maximum Temperature Analysis:

Maximum temperature analysis is done to comprehend the spatial patterns and strength of maximum temperatures between various districts during the time period in question. The data to be used in this analysis is based on ERA5 reanalysis data, which offers high resolution climate data. The data is then pre-processed and aggregated to get a maximum temperature value of each district averages which makes it possible to compare the variation of different regions. This study is mainly crucial in determining areas that are prone to extreme heat conditions and the patterns of temperature changes with time. The visualization methods like the bar graphs can be used so that the difference in temperature across the regions can be easily depicted and it becomes easy to analyze spatial differences.

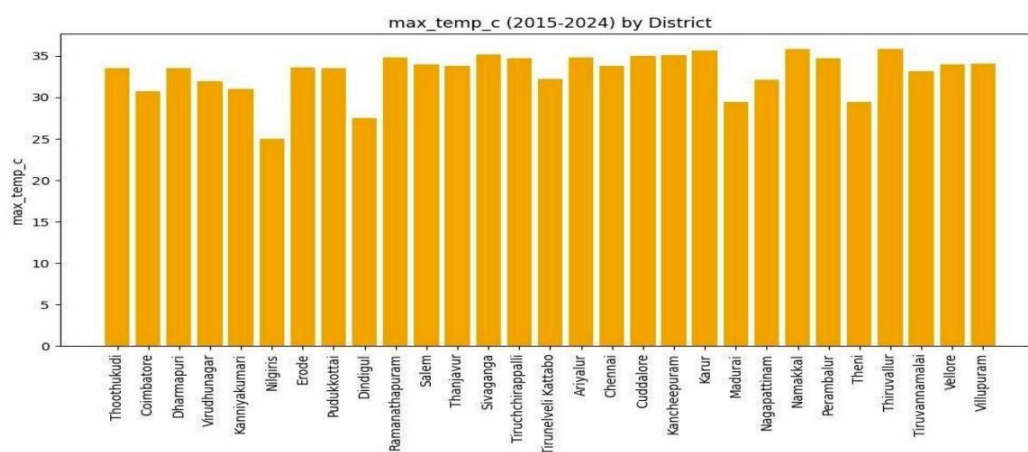


Fig 4.6 Max Temperature Analysis

Findings from Fig 4.6 indicate that some districts have persistent high values of maximum temperature than others, which implies that some areas are in a state of heat stress. This difference can be explained by the factors of urbanization, less vegetation cover, and geographical location. Areas where there is high development are more likely to retain more heat and therefore the temperature values are likely to be high, which is also known as the urban heat island effect. The observed high rise in maximum temperature of various locations is also in line with the general trends of the changes in climate, indicating a slow increase in the temperatures of the world and its regions.

(b) Minimum Temperature Analysis:

Minimum temperature analysis is concerned with the study of the temperature changes during the night and how this affects the overall stability in the climate. Minimum temperature gives information about the efficiency of the dissipation of heat at night, unlike the maximum temperature, which indicates daytime heat intensity. This parameter is highly important in the estimation of the long-term climate conditions since rising minimum temperatures tend to signify existence of more heat retention in the atmosphere. The data is treated like maximum temperature data and average per district is computed to provide the visualization of spatial distribution.

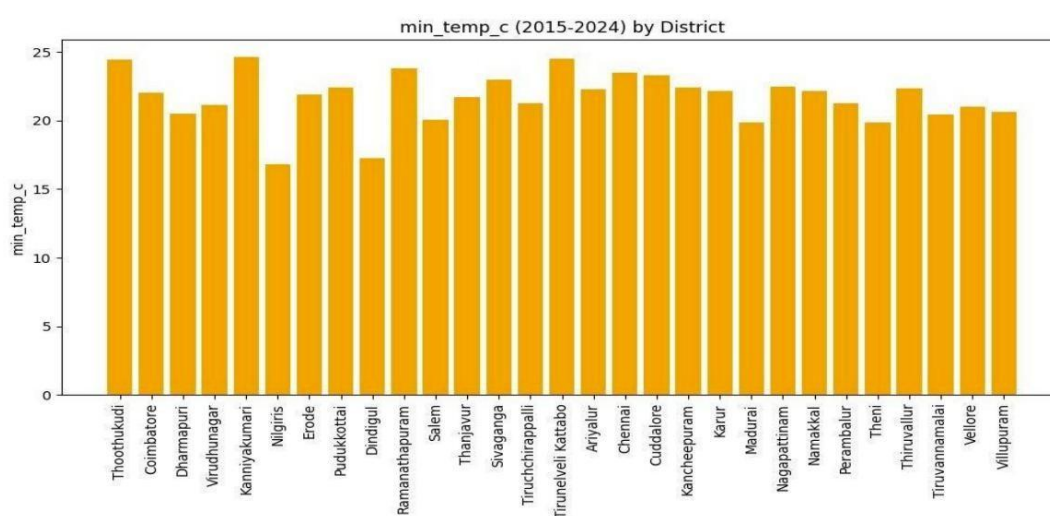


Fig 4.7: Min Temperature Analysis

The findings in Fig 4.7 reveal that there is a significant rise in the minimum temperature of most of the areas, implying that nights are getting warmer with the course of time. This is a good sign of climate change, as there is an accumulation of heat in the atmosphere which is indicated in this trend. The decreased day-night temperature variation may impact ecology and agriculture greatly, influencing crop growth and water needs. Also, the increased temperatures at night can result into increased energy use as they result in increased cooling needs. The uniformity of the trend in a variety of regions brings to focus the prevalence of fluctuating climate conditions. These observations show the necessity for adaptive strategies in land management to have long-term impact of temperature changes.

(c) Precipitation Analysis

The precipitation analysis is conducted where the distribution of the rainfall and variability are examined in different regions. The data set contains past data on the precipitation, which undergoes the model to obtain averages of the precipitation level of each district. This can be used to determine areas that get a lot of rainfall and others which are susceptible to drought. Precipitation patterns are needed in water resource management, agricultural planning, and preparedness to disasters.

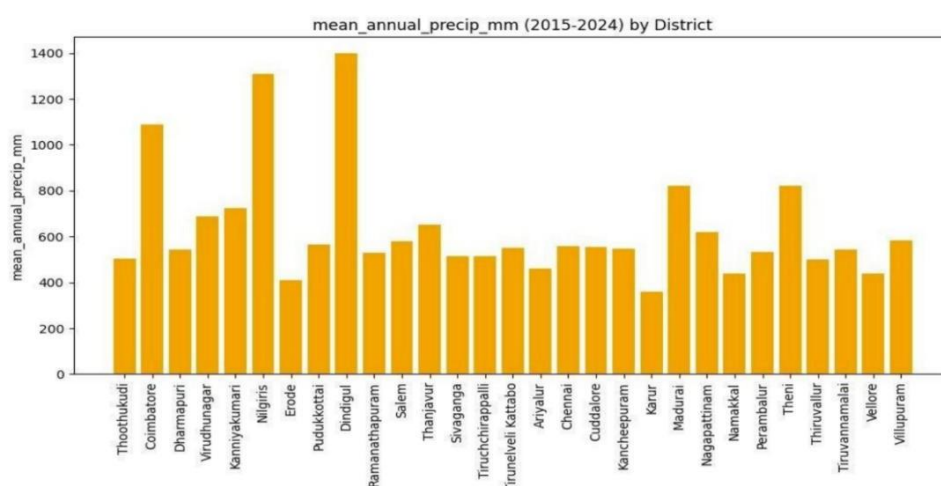


Fig 4.8: Precepitation Analysis

The findings as shown in Fig 4.8 indicate that there is a lot of differences in the distribution of rainfall with certain areas getting higher rainfall than others. Such imbalance suggests that growing uncertainty about weather patterns, which is a big attribute of climate change. Areas that experience higher rainfall levels are vulnerable to flooding and those with low level risk drought. The change of rainfall patterns also impacts agricultural productivity, where crops are greatly reliant on the availability of water. The results of these studies have highlighted the relevance of adaptive measures to deal with water resources and reduce the effects of climate change.

(d) Autoencoder Reconstruction

Climate data are analyzed with the autoencoder model that learns to compress representations of the input features and recreate the data. This will aid in detecting underlying trends and outliers in the data. The model is trained with historical climatic information, with it learning to encode the input into a smaller dimensional format where it then decodes it into its original form. The difference between the actual and reconstructed values gives information about the quality of the model in terms of reflecting the underlying data-structures.

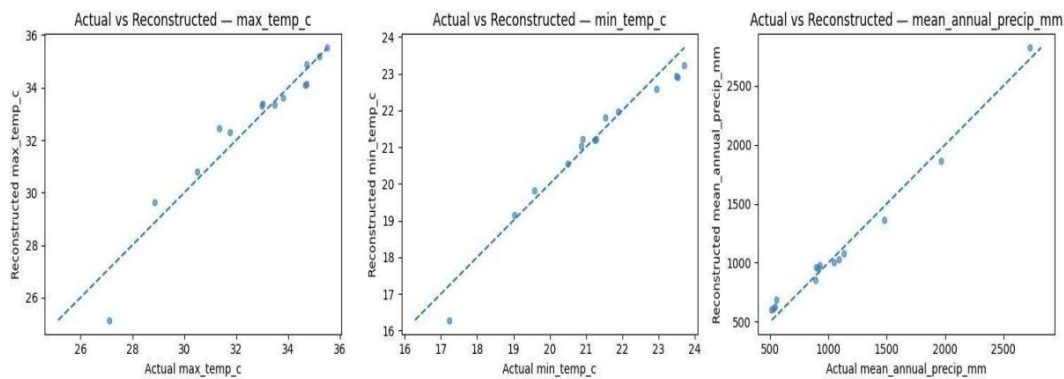


Fig 4.9: Actual vs Reconstructed results

Recent outcomes of the reconstruction shown in Fig 4.9 reveal that the autoencoder can closely reconstruct the original data, which means that it has learnt the underlying patterns in the climate data. Small differences between real and reconstructed values can be either anomaly or anomalous climate events, which can be of interest to future analysis. Such variations can be so readily detected using the autoencoder, thus making it a useful method of detecting anomalies in climatic studies. Generally, the findings verify that the unsupervised learning methods have the capability to extract intricate relationships in environmental data.

(e) LSTM Training and Performance Analysis

Long Short-Term Memory (LSTM) model is used to learn the temporal dynamics of climate data and make future predictions. Time-series analysis LSTM networks are well-adapted to predicting long sequences, As they support long-term memory with cells. Training of the model takes place with the training using historical climate data and validation using both training and validation loss curves. This enables the model to capture

long-term dependencies and complex temporal patterns in climate variables resulting in more accurate and reliable future predictions.



Fig 4.10 LSTM Loss Curve

The loss curve from Fig 4.10 exhibits a consistent drop in both training and validation loss, and it shows that the model is learning effectively and is generalizing well to unseen data. No major fluctuations indicate an unchanging convergence and little overfitting. This shows how LSTM networks can capture the temporal pattern of climate data with the use of time. The outcomes also show that the model is applicable in prediction of future climate patterns, hence it is a powerful tool in making predictions of the future. Generally, the effectiveness of the LSTM model justifies its use in modeling time-series in environmental research.

Similarly convergence of loss curve in Fig 4.10 is smooth indicating that the training process is stable and though the loss is greatly converted, the capability to recollect the long term dependence is still maintained by the model which is trained without much deterioration in the process.

4.1.3 RAG-Based System Results

(a) Semantic Similarity Evaluation using BERTScore

The semantic analysis of the suggested system is performed with the help of BERTScore, a context-sensitive evaluation measure, created to understand the semantic similarity between generated responses and reference answers.

In contrast to the conventional evaluation measures, e.g., BLEU or ROUGE, which are based on the presence of the exact matching of the words, BERTScore utilizes the use of transformer-based embeddings when the contextual meaning and word-word relationships are considered. It is therefore very applicable to systems which produce natural language output, especially in Retrieval-Augmented Generation (RAG) models where the emphasis lies upon natural language generation of answers that are accurate and relevant to the context. BERTScore is applied in this project to compare the performance of the baseline model, which produces answers without support via retrieval, and the suggested RAG-based model, which takes into account the appropriate context of the document and then produces the answers. The testing is carried out on various queries to allow consistency and reliability of outcome.

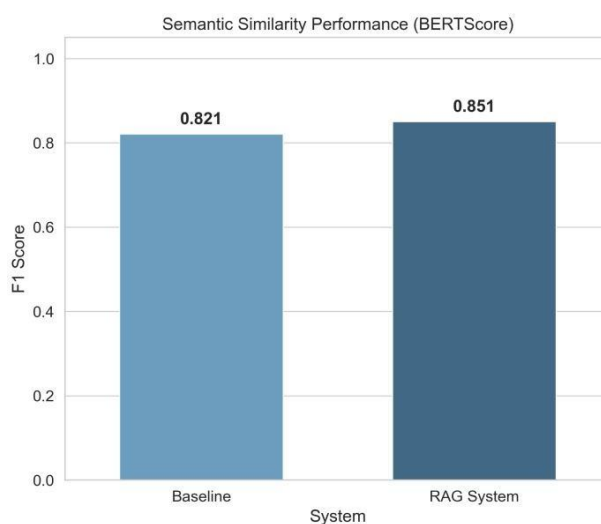


Fig 4.11 BERTScore Analysis

These findings from Fig 4.11, clearly show that the RAG-based system obtains a better BERTScore than that of the baseline model, which shows that there is a significant increase in semantic similarity. This enhancement shows the efficiency of integrating retrieval mechanisms in the process of response generation. The RAG system allows the language model to produce more source-relevant responses by supplying the model with contextual information pertinent to that document. This minimizes ambiguity and the quality of the answers generated is improved. The increased similarity of the semantics also means that the system has the capability of making out the meaning of the query and generates responses that are context-based instead of based on general or pre-trained information. This confirms

that retrieval augmentation is an important element of enhancing the performance of language models in document-based question answering.

(b) Lexical Similarity Evaluation with ROUGE-L

Besides the semantic evaluation, the system performance is also evaluated in terms of ROUGE-L that is a popular metric that has been employed to evaluate lexical similarity between generated and reference texts. ROUGE-L is preoccupied with the longest common occurrence between two text series and gives details about the extent to which the created answer upholds the structure and other significant elements of the reference answer. Although BERTscore encodes semantic content, the ROUGE-L measure encodes the extent of textual overlap thus becoming a significant supplementary measure of overall success. In this project, the ROUGE-L is employed to examine the effectiveness of the system to preserve structural consistency in the generated responses in the context of comparison with the baseline model.

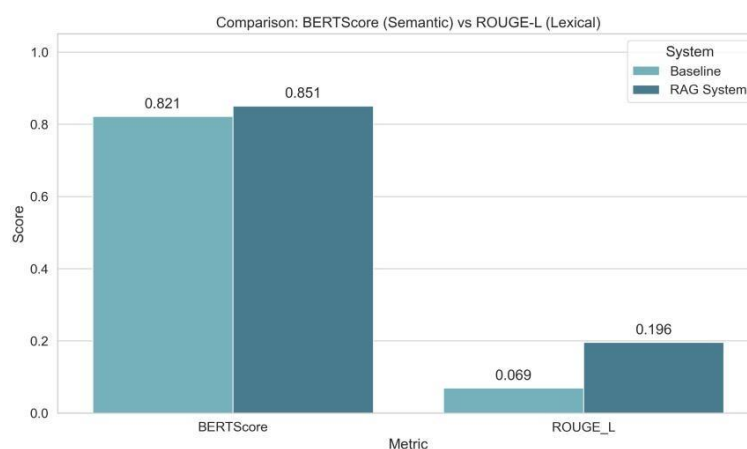


Fig 4.12 ROUGE Analysis

The findings shown in Fig 4.12 indicate that there has been a significant increase in the ROUGE-L scores of the RAG-based system over the baseline. This implies that the responses that are generated are not only capturing the correct meaning but also share a more structural similarity with the reference answers. The improvement in lexical similarity suggests that the system is able to include key terms and phrases from the source document, which enhances the clarity and relevance of the responses.

Though ROUGE-L scores tend to be lower than the values of BERTScore because of their stringent dependence on verbatim matches, the given improvement evidences that the RAG system attains the balance between the semantic understanding and textual accuracy.

(c) Combined Metric Analysis

BERTScore and ROUGE-L are evaluated together as metrics to obtain the full picture of the system performance. This combined analysis gives hints about the performance of the system both regarding semantic correctness and lexical similarity. Whereas BERTScore measures the meaning that is specific to the context, ROUGE-L measures structural alignment and the combination of the two metrics enables a more comprehensive evaluation of the application. This analogy is needed to comprehend whether the improvement in semantic accuracy will also lead to enhanced textual representation.

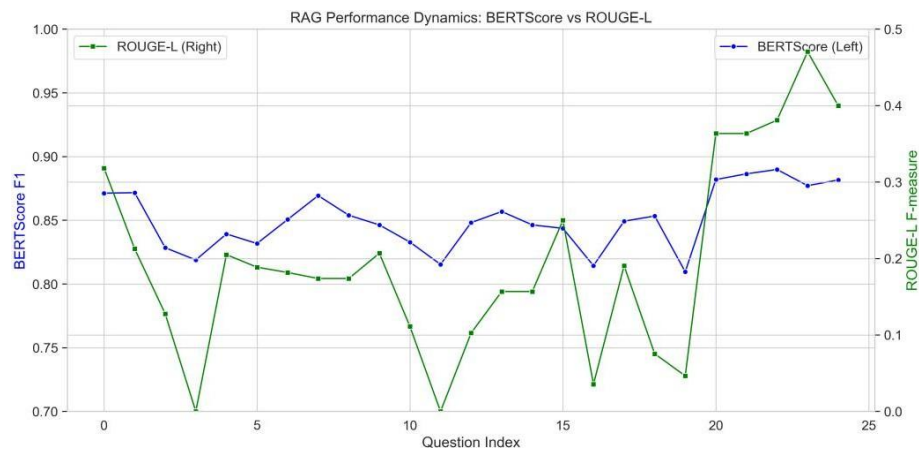


Fig 4.13 Combined Metric Analysis

The joint analysis (Fig 4.13) shows that the RAG-based system has a higher average performance compared to the baseline model on both of the metrics.

The difference between BERTScore and ROUGE-L indicates the disparity between semantic and lexical analysis, with BERTScore presenting higher scores by virtue of its context. Nevertheless, the fact that both measures increased shows that the system manages to increase the meaning and structure of the generated responses.

This proves that not only can the RAG approach provide contextually correct answers, but it also possesses a certain extent of textual coherence that can be linked to references. The

outcomes confirm the efficiency of the suggested methodology to deliver a balanced performance in various aspects of evaluation.

(d) LLM-as-a-Judge Evaluation

A qualitative analysis is also conducted in terms of the quality of responses generated, which is based on an LLM-as-a-judge method. In this approach, the responses generated are assessed by an independent language model on various criteria such as relevance, accuracy, completeness, and groundedness. This is a more humanistic approach of evaluation because it not only evaluates the correctness of the response, but also its quality and utility in general. The evaluation is carried out on both the baseline system and RAG-based system to compare their performance on the various qualitative dimensions

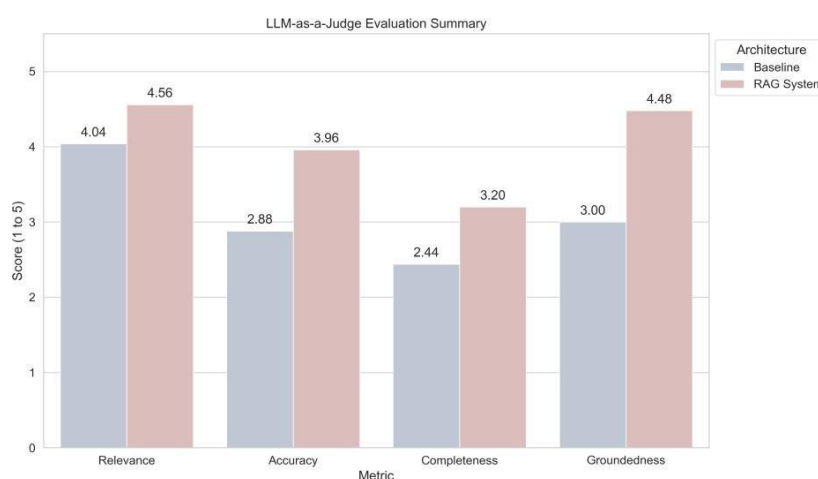


Fig 4.14 LLM-as-a-judge

As indicated by the outcomes of the LLM based evaluation in Fig 4.14, the RAG system scores much better on all the measures than the base model. Increasing relevance means that the system obtains responses that are well coherent to the query of the user. The larger accuracy score indicates the factual correctness of the answers and the increase in the completeness indicates that the system is more detailed and informative in its answers. The greatest change is seen in the groundedness which determines the extent to which the response is justified by the source document.

This proves the retrieval mechanism is efficient in reducing hallucination because responses used will be grounded on the content of the actual document. On the whole, these findings suggest that the RAG system yields more quality responses by various assessment standards.

(e) Analysis of Consistency and Stability.

Consistency analysis is carried out to verify the consistency of the system with several queries. This includes the study of the way the system behaves to various inputs, and whether it is consistent in the level of performance of the system. A stable system must give consistent results irrespective of the intricacy or fluctuation of queries. This discussion is especially significant when applied to real world models where the system has to deal with different and unpredictable inputs of the user.

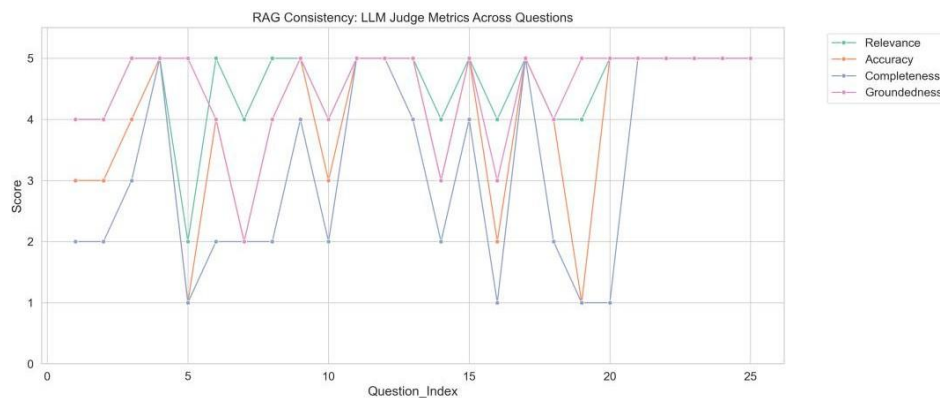


Fig 4.15 Consistency Analysis

The findings suggest that the RAG-based system (Fig 4.15) has a consistent performance on the various queries and the evaluation scores vary slightly. This stability goes to show the strength of the system and the capability it has to successfully accommodate any input situation. The homogeneity in the case of groundedness and relevance also proves the fact that the retrieval mechanism is a good contextual support that an individual should rely on in response generation. Conversely, the baseline model is more varied implying that the model outputs are less stable.

The new system will be developed in a way that will provide it with a clean, intuitive and easy-to-use interface to interface with the document analysis pipeline. Fig 4.16 shows the home page of the Academic Research Assistant that is the starting point of uploading the documents and starting analysis.

(f) Frontend UI Results

The landing page, which is in Fig 4.16 is minimal in design and structured so as to bring out usability and accessibility. The intent of the system is also presented in the interface i.e. to

streamline the scientific knowledge through the assistance of AI-generated idea of the documents. Get Started option provides the user with chances to work with workflow without any inconveniences and document upload option prepares the user with chances to work with academic PDFs.

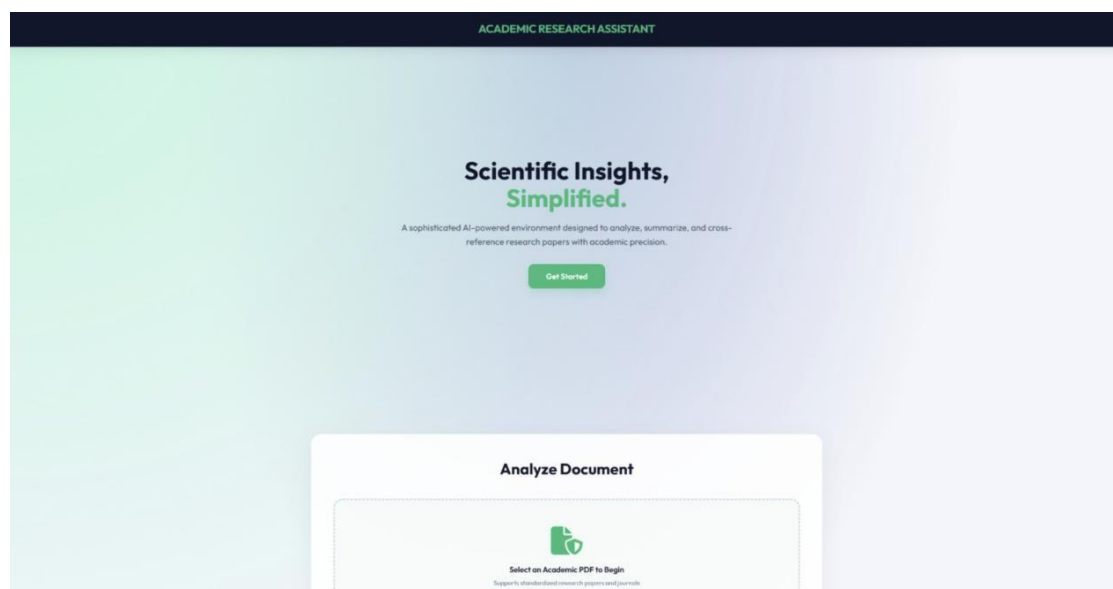


Fig 4.16 Landing Page

Fig 4.17 demonstrates the contextual inquiry interface, where the system processes user queries and generates responses grounded in the uploaded document content. The chat-based interaction model improves usability as it enables users to pose natural language questions, just like talking to people. The input it produces comprises summarized findings as well as supporting evidence that is retrieved out of the document, making it both transparent and reliable. Other important aspects that the system brings to light include the application of Retrieval-Augmented Generation, the accuracy of semantic retrieval and the limitations that are identified in the system including hallucination and bias. The interface successfully fills the gap between sophisticated document analysis and user-friendly interface to extract knowledge efficiently. Furthermore, the inclusion of citation-like references within the response enhances trustworthiness and makes the system suitable for academic and research-oriented applications.

Contextual Inquiry

What use case does the paper explore?

The paper investigates using Retrieval-Augmented Generation to answer queries from a PDF-based knowledge base.

- Applies RAG with a proprietary collection of PDF documents.
- Focuses on PDF-based content retrieval and semantic retrieval accuracy.
- Discusses limitations such as hallucination, bias, and plagiarism.

[1] INTRODUCTION: "2 Authorized licensed use limited to: Rajshahi University Of Engineering and Technology. Downloaded on April 03,2026 at 06:55:25 UTC from IEEE Xplore. Restrictions apply. TABLE I: Technologies Used ..."

[2] INTRODUCTION: "from a proprietary knowledge base made of a PDF file and"

[3] Vectorization and Storage: "A. PDFs-based Content Retrieval 1) Semantic Retrieval Accuracy: Table II, represents the"

[4] INTRODUCTION: ". The paper also examined key limitations, however, such as the creation of misinformation ("hallucination"), the reinforcement of biases from imperfect training data, and ethical concerns like plagiarism..."

Ask a technical question about the paper...

Send

Reset

Fig 4.17 Chat Box

4.2 MODELS PERFORMANCE COMPARISON

4.2.1 Comparison of Models on LULC Classification

LULC Classification mostly compares the performance in terms of classification accuracy and reliability of models being used to analyze Land Use and Land Cover (LULC). The two models, under evaluation, Support Vector machine (SVM) and Random Forest (RF), were tested using the same datasets and preprocessing steps to compare using the same method. Based on the findings in the above section, it is clear that both models can be effectively used in classification works, but they perform in a significantly different way when it comes to complex and heterogeneous land cover patterns. SVM, which works by determining an optimal hyperplane to classify the data, works well with data that are linearly separable, as well as moderately complex. Nevertheless, the performance of SVM would tend to reduce when land cover classes share similar spectral properties, which is more fit to handle high-dimensional and noisy data. The findings clearly indicate that in all the timelines being considered, Random Forest has better classification accuracy and a stronger separation of the classes than SVM. Also, Random Forest performs better in classifying similar land cover

types like agricultural and fallow land that in many cases are tricky to separate using more traditional approaches. This comparison shows the significance of choosing the right machine learning models in geospatial analysis and validates that ensemble-based models are more effective with large datasets to classify.

4.2.2 Comparison of Models for Climate Analysis

In this step, performance comparison will be done on deep learning models applied to analyse climate data specifically Autoencoders and Long Short Term memory (LSTM) network.

These models aim to model intricate patterns and time dependence in data and thus are appropriate in analyzing climate variables like temperature and precipitation. The autoencoder model is tested according to the possibility to reconstructing the input data, and the LSTM model is tested according to the possibility to learn the temporal dynamics and predict the future values. Based on the findings of the above section, it is clear that both models are performing well, albeit with minor differences in their roles and strengths.

The autoencoder model is highly effective in the reconstruction and detection of anomaly in underlying data structures as indicated in the high reconstruction accuracy. It is especially useful in unsupervised learning processes that are not provided with labeled data. Conversely,

The LSTM model has better performance in capturing temporal dependencies as it has a consistent convergence and can learn long-term trends in climate data. The declining trend of loss experienced in training confirms the fact that the LSTM model is an effective model in capturing sequential information. Comparatively, the auto-encoder suits feature extraction and anomaly detection more than the LSTM that is suited better in prediction and trend analysis. This is the symbiotic relation between the two models that emphasize the need to employ various methods to get a broad analysis in climate studies.

Moreover, the complementary nature of Autoencoders and LSTM models underscores the significance of combining various deep learning models in order to analyze climate data comprehensively. Although Autoencoders can be useful in capturing underlying data distribution and detecting anomalies, LSTM networks have powerful capabilities in modeling of temporal sequence and predicting future trends. This combination allows seeing climate behavior more holistically, as it tackles both structural patterns and variations over time. This type of integrated approach makes the analysis more robust and reliable, and suitable to both real-world environmental monitoring and predictive applications.

4.2.3 Comparison of Models for RAG and Baseline LLM

This project presents a Retrieval-Augmented Generation (RAG) framework, which is contrasted with a baseline large language model, which does not have retrieval assistance. It is compared both quantitatively based on BERTScore and ROUGE-L and qualitatively based on the metrics of an LLM-as-a-judge. The baseline model is only based on already trained knowledge to produce responses, which can be problematic in other aspects, including hallucination and lack of context. RAG system, on the contrary, has a retrieval system that gives the context of a relevant document.

Table 4.1 presents the overall comparison of the performance of the proposed RAG system with the baseline LLM on various evaluation metrics.

Table 4.1 Overall System Performance

Metric Type	Metric	Baseline (Llama 3.3 70B)	RAG System	Improvement (%)
NLP similarity	BERTScore (F1)	0.8215	0.8508	+3.57%
	ROUGE-1	0.0907	0.2204	+143.00%
	ROUGE-L	0.0692	0.1957	+182.80%
Human-centric	Relevance (1-5)	4.04	4.56	+12.87%
	Accuracy (1-5)	2.88	3.96	+37.50%
	Completeness (1-5)	2.44	3.20	+31.15%
	Groundedness (1-5)	3.00	4.48	+49.33%

As seen in Table 4.1, the RAG system is steadily improving in terms of semantic as well as human-centric evaluation metrics. The rise in BERTScore corresponds to a better contextual comprehension whereas the increase in ROUGE measures implies a more accurate structural correspondence with reference answers. The large increase of groundedness and accuracy point out the fact that the system can minimize hallucination and produce information which is factual.

Based on the findings above, it is clear that the RAG system is much better than the baseline model in all the evaluation metrics. These changes in BERTScore and ROUGE-L demonstrate better semantic and structural alignment with reference answers, respectively. Additionally, the qualitative assessment results indicate more points in relevance, accuracy,

completeness, and groundedness, which supports the fact that the RAG system creates more accurate and contextually relevant answers.

Among the key benefits of the RAG approach is that they help minimize hallucination by basing responses on real document content. This renders the system more appropriate in real world applications where accuracy and reliability is essential. Table 4.2 gives a breakdown of the lexical and semantic evaluation measures in detail.

Table 4.2 Lexical and Semantic evaluation

System Configuration	BERTScore	ROUGE-1	ROUGE-2	ROUGE-L
Llama 3.3 70B	0.8215	0.0907	0.0224	0.0692
RAG System	0.8508	0.2204	0.0979	0.1957
Absolute Gain	+0.0293	+0.01297	+0.0755	+0.1265

Table 4.2 also supports the idea that the RAG system increases the semantic similarity and lexical alignment. The recorded benefits evidenced that integration of retrieval not only improves the meaning of generated responses but also their structural integrity. Besides quantitative analysis, Table 4.3 reports qualitative analysis based on the metrics of LLM-as-a-judge.

The qualitative assessment also reinforces the findings of the quantitative measures by giving an understanding of the performance of the built system with respect to the human-based assessment. The results of higher scores in relevance, accuracy, completeness, and groundedness mean that RAG system produces not only factually correct responses but also makes sure that information produced is contextually relevant and detailed enough. Specifically, the enhancement of groundedness is noticable and the capability of the system to anchor its own responses on the content of recalled documents, and thus decrease the occurrence of hallucinations and increase reliability. This combined assessment technique shows that the suggested method will provide a balanced score improvement on both quantitative measures of performance and real-life cases, which will make it more effective in practice applications to document-grounded question answering assignments.

Table 4.3 LLM-as-a-judge

Metrics	Llama 3.3 70B	RAG System	Gain
Relevance	4.04	4.56	+0.52
Accuracy	2.88	3.96	+1.08
Completeness	2.44	3.20	+0.76
Groundedness	3.00	4.48	+1.48

Table 4.3 has given a clear indication that the RAG system is superior to the one being under the baseline model in all the qualitative aspects. The increases in relevance and completeness demonstrate that the system offers more meaningful and detailed answers, whereas the tremendous rise in groundedness proves that the answers generated by the system are highly supported by contents of retrieved documents. The proposed system is more reliable and can be used in practice because of this decrease in hallucination. Overall, this table shows that RAG systems are more suitable for document-grounded answering.

4.2.4 Overall Comparative Analysis

The general comparison of all the projects outlines the development of the complexity and performance of models of traditional machine learning to deep learning and ultimately to sophisticated hybrid architectures. In LULC Study, classification is performed using machine learning models that are of classical form, and the ensemble-based approach like a random forest is better than the simplistic models like SVM. Deep learning models used in Climate Analysis are able to process more detailed and time-dependent data, which proves the superiority of neural networks in comparison with conventional methods. This development is further enhanced by the RAG Framework in that it incorporates the retrieval mechanisms with the generative models leading to the creation of a hybrid system that incorporates the strengths of retrieval and generation

This development indicates clearly that, the more complex the problem, the higher the level of sophistication the model should have to perform well. The shift toward SVM and then Random Forest and traditional models to deep learning and RAG-based systems are indicative of the increasing demand of large-scale, complex, and unstructured data models. All the findings show that AI systems that use a combination of several approaches are more effective to handle real-world issues in the modern world.

It is also evident through the relative development of the three projects that adaptability, scalability and the capability to be conscious of the context are relevant with contemporary intelligent systems. Despite the practicality of the conventional machine learning models in case of structured and defined data, its flaws are evident with high-dimensional and unstructured data with context. Deep learning models fill this gap, that is, they model the complex patterns and time dependence, but, unfortunately, do not have any relationship with reality and external information in real-time. The RAG-based architecture is able to overcome this shortcoming in the sense that it can experience dynamic retrieval with generative properties in that they can be factual and contextual in the responses. This development, besides being a technological breakthrough, shows that any intelligent systems in the future should be hybrid in nature, having several methodologies to enable a higher degree of accuracy, reliability and wide applicability to the most diverse real world situations.

On the whole, the overall comparison, with all of three studies, is a clear indication of a gradual move to more intelligent, adaptive, and context-aware systems, which are able to take into consideration complex real-world problems. This fact is reflected in the transition of classical machine learning to deep learning and, later, to hybrid RAG-based designs, which highlights the necessity to combine the different approaches in order to achieve optimal performance. The development is not only more precise and effective, but this expansion also raises the potential of the system in unstructured information processing and yielding valid and context-based outputs. The results, therefore, support the fact that more innovations in artificial intelligence will be based on combined and hybrid solutions, which will guarantee scalability, robustness, and practical implementation in various applications.

4.3 OBSERVATIONS FROM WORK

Changes in the utilization of land can be observed throughout the study time - urban sprawl is evident, with individuals transforming space. Rather than plots of land or forested areas, concrete is poured, where crops or trees used to grow. These changes are not motivated by numbers alone but by roads and buildings as well; this indicates underlying pressures on space. Natural systems become smaller and this is an indicator of damage to the natural systems as well as potential extinction of species. In the battle of the algorithms, one of them wins, not because it is simpler but because it has a stratified way of picking up hidden data patterns. Random Forest tends to perform better due to its complexity, because it is able to work with curved data shapes and has low levels of false precision, whereas SVM is unable to work well in landscapes that contain sharp contrasts of types. These disparities underscore

the importance of model choice in the research of ecosystems that are directly related to the ways in which people transform the landscape.

In all the regions surveyed, there are indications of changing climates and increasing environmental imbalance - data obtained first-hand in the course of Climate Analysis. Warming is evident with consistent increases in both peak and low temperatures, consistent with larger global patterns. It is interesting to note that lows at night are increasing faster, indicating that the air could be retaining heat longer, which decelerates the restoration of heat at night. This change can silently transform ecosystems and disrupt crop production over the years. Rainfall patterns take on an extra dimension: they do not come on time, they are not evenly distributed, they are even directed at untrustworthy seasonal changes. The unpredictable changes pose a challenge in controlling the water sources coupled with the agricultural patterns, as dry seasons and heavy rains occur more frequently. Turning to computational approaches, autoencoders combined with LSTM networks can show how deep learning can deal with complex time-varying patterns. The autoencoder can reveal hidden data shapes, which may be imperceptible to others. In the meantime, the LSTM carries along the changing sequences throughout a time span, which is useful in the prediction of future states. These outcomes suggest that intelligent analysis tools have further applications to use in addressing climate-related problems.

The interesting detail that can be distinguished against the primary focus is that Retrieval-Augmented Generation can increase the document comprehension and response quality in big language systems. One fundamental lesson is that responses are much more accurate in their semantics and contextual appropriateness once the elements of retrieval are added to the process. These upgraded models do not rely solely on stored training information, instead of including external documents in their response formation. Due to this change, fabricated information falls out of favor - something that is often a weakness of typical systems that do not have access to real evidence. The most remarkable aspect is that the system is stable regardless of the type of query - this indicates high adaptability in different situations. Such performance metrics as BERTScore, ROUGE-L, and even expert-like reviews by larger models demonstrate evident benefits compared to conventional approaches. Alternatively to search using keywords, meaning-based search using vectors enables better understanding of complex questions and accelerates the process of searching data. Since both extraction of facts and development of answers can be very effective in combination, this combination is potent in terms of creating viable AI tools that can be expanded and sustained.

4.4 SUMMARY

From the above study, this section reviewed outcomes from LULC Classification, Climate analysis, and the RAG pipeline, showing how various AI methods perform across diverse areas. Though simpler algorithms were used at first, later steps revealed stronger patterns when deeper networks took over. Random Forest along with Support Vector Machine handled land-based classifications well, especially where spatial boundaries mattered. Where time played a role, LSTM and autoencoders captured subtle shifts within climate records others missed. As problems grew in difficulty, basic tools gave way to systems better suited for layered data. Each task brought something distinct - tracking ecosystems, decoding weather trends, organizing documents intelligently. Performance gains appeared only once model depth matched challenge scale. What stood out was not just accuracy but consistency under changing conditions. Earlier stages relied on structured inputs; later ones adapted without fixed rules. The takeaway emerges quietly: method choice shapes outcome far more than effort alone.

One key outcome stood out - Retrieval-Augmented Generation boosted how well big language models answer questions using documents. Instead of relying solely on pre-trained knowledge, pulling in relevant context before generating answers led to sharper, more accurate replies. Because information was retrieved first, made-up details appeared far less often. When looking at every project together, movement toward blended methods became obvious over time. Findings back up the core approach taken here. Progress now rests on this base - ready to grow into systems that handle tough tasks reliably.

CHAPTER-5

CONCLUSION AND FUTURE WORK

This chapter gives the end of the implementation and assessments on the suggested Retrieval-Augmented Generation (RAG)-based system of reading and responding to user questions in the documents. It gives the overview of the key findings of the system and demonstrates its effectiveness in increasing the accuracy of responses and its topicality to the context. Besides, the chapter has also outlined potential future work to the system to improve it and add to its potential value in the real-world use cases.

Another key feature discussed in the chapter is that the combination of retrieval processes and generative processes allows the system to address the major drawbacks of the traditional methods, especially regarding the facts and context. The system proves to be able to produce credible and significant answers by showing the steady improvement of various performance measures through thorough evaluation and analysis. The findings highlight the practical utility of the presented framework in processing complex and unstructured documents, as well as the scalability and flexibility of the framework to implement it in the real-life context.

5.1 CONCLUSION

The success of the combination of retrieval mechanisms with large language models to solve document-based questions is effectively presented in the proposed system. It can be very efficient with large unstructured documents as it integrates semantic chunking, embedding generation and storing the information in vectors and retrieving the information by the use of cosine similarity. LLaMA 3.3 70B then applies this retrieved context to generate accurate, meaningful and well-structured outputs and reduces hallucination and ensures the output is informed by real data.

The Retrieval-Augmented Generation (RAG) architecture has increased reliability and factual accuracy to a large extent because it closely combines retrieval and generation. The evaluation findings suggest clear improvements over the baseline models, in both metrics (BERTScore, ROUGE-L) and qualitative (relevance, completeness, groundedness) performance. This proves that the system does not just give out the right answers, but also provides context-enhanced and detailed answers applicable in applications where high accuracy is warranted.

In addition, the system is scalable, flexible and has the ability of managing long and complex documents through overcoming context length limitations by semantic chunking and semantic retrieval. Its scalability to other areas and types of queries ensures that it is scalable to real life applications such as research, technical analysis and knowledge management. In total, the proposed solution provides a solid, legitimate, and multi-purpose paradigm of intelligent document recognition and reveals the potential of the RAG systems within the context of AI-based application in the future.

5.2 FUTURE WORK

Future efforts of this system could be directed on the enhancement of the efficiency of retrieval as well as the overall scalability. The retrieval techniques currently in place are based on semantic and keyword based retrieval, which can be enhanced by integrating both techniques in hybrid methods. This will assist in retrieving more pertinent parts of documents. Moreover, exploring improved chunking schemes and more advanced embedding models can lead to improved accuracy and also lead to lower computational cost. It is also possible to expand the system to enable multi-document retrieval to enable it to process bigger datasets and provide more complete responses in real-world situations.

The other significant direction is to enhance the quality and intelligence of the generated responses. This can be done by conditioning large language models with domain-specific data like legal, medical, or technical data. This will assist the system to be able to better comprehend context within special domains. The addition of user feedback and reinforcement learning methods can also help to make further improvements and keep improving the performance. In addition, transparency and trust can be enhanced using better factual accuracy evaluation methods and explainable AI techniques. The system could be made more robust and applicable to real-world scenarios through additions such as model compression to make it efficient and the ability to handle multimodal inputs such as text, images, and tables.

Systemwise, future work can also be aimed at enhancing deployment, usability and real-time performance. The use of cloud-based or distributed architectures will enable the system to manage a vast amount of data and multiple users effectively. Caching, parallel retrieval and efficient indexing are some of the techniques that can also be used to optimize performance to reduce the response time in real-time applications. Besides, the user interface can be enhanced to make it more interactive and adaptive and contribute to the overall user

experience. The system can be made easier to use with features like personalized responses, suggestion of queries and enhancement of visualization of information that is retrieved.

Lastly, the proper security and privacy mechanisms must be included, particularly when working with sensitive documents. Data protection and safe processing will be extremely important in the real-world adoption. Through these enhancements the system can become a research prototype, a full-fledged deployable and production-ready intelligent document analysis system.

REFERENCES

- [1] P. Santiago García-Montero, P. Vizcaíno, I. Galo Reyes-Chacón and M. Eugenio Morocho-Cayamcela, “Legal AI for All: Reducing Perplexity and Boosting Accuracy in Normative Texts With Fine-Tuned LLMs and RAG,” *IEEE Access*, 2025, doi: 10.1109/ACCESS.2025.3622138.
- [2] H. Tekgöz, H. Erdoğan, H. Uz and H. Altınok, “Evaluation of Vector Database and LLM Models in Retrieval-Augmented Generation (RAG) Systems,” 10th International Conference on Computer Science and Engineering (UBMK), Istanbul, Turkey, 2025.
- [3] B. Saha, U. Saha and M. Z. Malik, “QuIM-RAG: Advancing Retrieval-Augmented Generation With Inverted Question Matching for Enhanced QA Performance,” *IEEE Access*, 2024.
- [4] G. Regina Elwin J, R. Narayanan B, R. Rithik Raj K. S and M. R., “AI-Driven Solution for Analyzing and Interpreting Data from PDFs and Other Textual Resources Using RAG and Hybrid Retrieval Models,” International Conference on Research Methodologies in Knowledge Management, Artificial Intelligence and Telecommunication Engineering (RMKMATE), Chennai, India, 2025.
- [5] A. Sonkar, A. Sahu, S. P. Singh, S. Mishra and K. Sahu, “Dynamic Query Handling with RAG Fusion for PDF-Based Knowledge Retrieval Systems,” 4th OPJU International Technology Conference (OTCON), Raigarh, India, 2025.
- [6] H. Shiralaskar, S. Dagadkar, K. Wanjari, S. S. Chavhan, T. Gawande and H. Chamatkar, “The State of the Art in Conversational Document Analysis: A Review of RAG-Based Architectures,” IDICAIHEI Conference, 2025.
- [7] T. Kolla, T. Babu and R. R. Nair, “Evaluating Hallucinations in Large Language Models (LLM): Metrics and Mitigation,” *IEEE INDISCON Conference*, 2025.
- [8] S. Shaikh and A. Halbe, “Dual-Source Retrieval for Educational Chatbots: A Performance Evaluation of Semantic Embedding Models,” International Conference on Intelligent Systems and Applications, 2025.
- [9] S. Suryawanshi, S. Waghmode, R. Sawant, and M. Gupta, “A Knowledge Graph-based RAG for Cross-Document Information Extraction,” in *Proc. 5th Int. Conf. on Pervasive Computing and Social Networking (ICPCSN)*, 2025, pp. 1401–1406, doi: 10.1109/ICPCSN65854.2025.11035854.
- [10] V. Gogineni, “LLM-Powered Multi-Agent Systems: A Technical Framework for Collaborative Intelligence through Optimized Knowledge Retrieval and Communication,” *AIRC Conference*, 2025.
- [11] M. Niinimäki, S. R. Shrestha and A. Udofia, “Comparison of Vector Database Management Systems for Retrieval Augmented Generation,” International Computer Science and Engineering Conference (ICSEC), 2025.

- [12] A. M. S and K. S. Easwarakumar, “Beyond ROUGE for Legal-Domain-Specific Evaluation Metrics for Judicial Text Summarization,” RMKMATE Conference, Chennai, India, 2025.
- [13] I. Ni’Mah, L. R. Aini, R. Fajri, S. Pebiana, N. N. Hidayati and R. Wijayanti, “Evaluating Retrieval Augmented Generation (RAG) Chunking Strategy for Question Answering in Indonesian Law of The Sea,” International Conference on Computer, Control, Informatics and its Applications (IC3INA), Jakarta, Indonesia, 2025.
- [14] R. More, “A Unified Evaluation Framework for Grounded LLM Architectures: Comparative Analysis of RAG, Self-RAG, and Agentic RAG,” International Conference on Artificial Intelligence and Signal Processing (AISP), Vijayawada, India, 2025.
- [15] M. Aguirre et al., “Centinela Meets Agents: Agentic RAG for Academic Information in Ecuador,” 2025 LI Latin American Computer Conference (CLEI), Valparaíso, Chile, 2025, pp. 1–4, doi: 10.1109/CLEI67442.2025.11420412.
- [16] J. Yu et al., “MIND-RAG: Multimodal Context-Aware and Intent-Aware Retrieval-Augmented Generation for Educational Publications,” 2025 IEEE/CVF International Conference on Computer Vision Workshops (ICCVW), Honolulu, HI, USA, 2025, pp. 4216–4223, doi: 10.1109/ICCVW69036.2025.00438.
- [17] I. Mahmood, M. Mastan, Z. Naveed, H. Iftikhar, W. Ahmad and S. M. Adnan, “AzureIQ-RAG: Hybrid RAG System for Automated Information Retrieval and Semantic Query Answering Using Azure AI,” 2025 12th International Conference on Information Technology (ICIT), Amman, Jordan, 2025, pp. 577–583, doi: 10.1109/ICIT64950.2025.11049144.
- [18] S. Mehta, T. Negandhi and S. Ghane, “Finalyze: A RAG-Based Framework for Intelligent Financial Document Analysis,” 2025 5th International Conference on Emerging Research in Electronics, Computer Science and Technology (ICERECT), Mandya, India, 2025, pp. 1–5, doi: 10.1109/ICERECT65215.2025.11377885.
- [19] N. Wang, X. Zhao, X. Lin, X. Liu and Q. Sun, “Retrieval-Augmented Generation for Biomedical Question Answering Based on Hierarchical Knowledge Tree,” 2025 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), Wuhan, China, 2025, pp. 7438–7444, doi: 10.1109/BIBM66473.2025.11356417.
- [20] H. S. Rosari, Girinoto, R. Novita Yasa, N. Qomariasih and R. Anindya Wijayanti, “SiJesi: Large Language Model Chatbot with Augmented Retrieval Approach Generation and Prompt Engineering,” 2024 International Conference on Computer Engineering, Network, and Intelligent Multimedia (CENIM), Surabaya, Indonesia, 2024, pp. 1–6, doi: 10.1109/CENIM64038.2024.108

APPENDIX

The development of the Research Paper Analysis system focused on building a robust Retrieval-Augmented Generation (RAG) pipeline tailored for academic accuracy. The architecture prioritizes strict contextual grounding and automated metric-based evaluation to ensure high-quality, verifiable summaries. The following appendices detail the underlying logic for the system's prompting, performance benchmarking, and document parsing infrastructure.

Fig A1 illustrates the prompt engineering strategy used to enforce a "Strict Research Assistant" persona. It defines rigorous constraints on output length and format, ensuring that the model provides concise, bulleted facts while explicitly preventing hallucinations by requiring a "don't know" response if the context is insufficient.

```
prompt_template = """
You are a strict research assistant. Answer the question using ONLY the
provided context.

Context:
{context}

Question:
{question}

Instructions:
- Summarize the answer in your own words. DO NOT exactly copy sentences.
- STRICT LENGTH LIMIT: Your total response MUST be under 350 characters.
- FORMAT: Start with a brief 1-sentence summary. Then provide 2-3 short
bullet points.
- No fluff or extra jargon. Give only the core facts.

If the answer is not clearly present, say: "I don't know."
"""
```

Fig A1: RAG Brain

Fig A2 showcases the evaluation framework used to automate the benchmarking of system responses. By defining a specific "Judge" prompt, the system scores generated answers on a 1-5 scale across four key metrics—relevance, accuracy, completeness, and groundedness—providing structured JSON data for statistical performance analysis.

```
# -----  
# LLM-as-a-Judge Logic  
# -----  
JUDGE_PROMPT_TEMPLATE = """  
You are an expert evaluator. Assess the following AI-generated answer.  
Compare it strictly to the provided Reference Answer and Source Context.  
Score each metric from 1 to 5. Output ONLY valid JSON, do not include markdown blocks.  
  
Metrics:  
- relevance: Does the AI answer directly address the Question?  
- accuracy: Is the AI answer factually correct according to the Reference Answer?  
- completeness: Does the AI answer cover all essential points from the Reference Answer?  
- groundedness: Is the AI answer supported by the Source Context? (CRITICAL EXCEPTION: If the  
  
Question: {question}  
Reference Answer: {reference}  
Source Context: {context}  
AI Answer: {answer}  
  
Required JSON Output Format:  
{ "relevance": <int>, "accuracy": <int>, "completeness": <int>, "groundedness": <int> }  
"""  
  
judge_prompt = PromptTemplate(template=JUDGE_PROMPT_TEMPLATE, input_variables=["question", "re
```

Fig A2: LLM-as-a-judge Evaluation Logic

Fig A3 depicts the logic used to transform raw OCR-extracted text into a structured, section-aware format. This function intelligently splits research papers into distinct sections and subsections, allowing the retrieval engine to index data more granularly and point users to specific headers within the source material.

```
def split_sections_with_content(text: str, detected_sections: List[Dict]) -> List[Dict]:
    """
    Split text into sections/subsections using detected start positions.
    Returns a List of dicts with: section, subsection (if any), start, content.
    """
    if not detected_sections:
        return {"Full_Paper" : text}

    # Sort by start index to ensure correct order
    detected_sections = sorted(detected_sections, key=lambda x: x["start"])
    results = {}

    for i, sec in enumerate(detected_sections):
        start = sec["start"]
        end = detected_sections[i + 1]["start"] if i + 1 < len(detected_sections) else len(text)

        # Section details
        section_name = sec["section"]
        subsection_name = sec.get("subsection", None)
        section_text = text[start:end].strip()

        results[section_name] = section_text

        if subsection_name:
            results[section_name][subsection_name] = section_text

    return results
```

Fig A3: Section Splitting Logic

Fig A4 demonstrates the final stage of the RAG loop, where retrieved evidence is deduplicated and formatted for the web interface. It ensures that every response is accompanied by clear source citations, including section names and content snippets, to maintain transparency and facilitate user verification

```
source_texts = []
seen_content = set()

for doc in sources:
    content = doc.page_content.strip().replace("\n", " ")
    # Avoid showing the same text twice
    if content[:50] in seen_content:
        continue
    seen_content.add(content[:50])

    section = doc.metadata.get("section", "Context")
    # Show a meaningful snippet
    snippet = content[:200] + "..." if len(content) > 200 else content
    source_texts.append(f"[{len(source_texts)+1}] {section}: \"{snippet}\"")

# Return separately so frontend can style them
return jsonify({
    "response": answer,
    "sources": "\n".join(source_texts) if source_texts else ""
})
```

Fig A4: Source Citation Logic

10% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text
- Crossref database
- Crossref posted content database

Match Groups

-  **186 Not Cited or Quoted 10%**
Matches with neither in-text citation nor quotation marks
-  **0 Missing Quotations 0%**
Matches that are still very similar to source material
-  **0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 3%  Internet sources
- 2%  Publications
- 6%  Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Disclaimer

Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?

The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?

Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

